15. Management of Diabetes in Pregnancy: Standards of Care in Diabetes—2026

*American Diabetes Association Professional Practice Committee for Diabetes**

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DIABETES IN PREGNANCY

The prevalence of diabetes in pregnancy has been increasing in the U.S. in parallel with the worldwide epidemic of obesity. Not only is the prevalence of type 1 diabetes and type 2 diabetes increasing in individuals of reproductive age but there is also a dramatic increase in the reported rates of gestational diabetes mellitus (GDM). Diabetes confers significantly greater maternal and fetal risk that is largely related to the degree of hyperglycemia but also is related to chronic complications and comorbidities of diabetes. In general, specific risks of diabetes in pregnancy include spontaneous abortion, fetal anomalies, preeclampsia, fetal demise, macrosomia, neonatal hypoglycemia, neonatal hyperbilirubinemia, and neonatal respiratory distress syndrome. In addition, exposure to hyperglycemia in utero increases the risks of obesity, hypertension, and type 2 diabetes in offspring later in life (1,2).

Preconception Counseling

Recommendations

15.1 Starting at puberty and continuing in all people with diabetes and childbearing potential, preconception counseling should be incorporated into routine diabetes care. **A**

15.2 Family planning should be discussed, and effective contraception (with consideration of long-acting, reversible contraception) should be prescribed and used until an individual's treatment plan and A1C are optimized for pregnancy. **A**

15.3 Preconception counseling should address the importance of achieving glucose levels as close to normal as is safely possible without excessive hypoglycemia, ideally A1C <6.5% (<48 mmol/mol), to reduce the risk of congenital anomalies, preeclampsia, macrosomia, preterm birth, and other complications. **A**

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15.4 Individuals with a history of gestational diabetes mellitus (GDM) should seek preconception screening for diabetes and preconception care to identify and treat hyperglycemia and prevent congenital malformations and other adverse maternal and fetal outcomes. **E**

Preconception counseling for pregnant people with preexisting type 1 or type 2 diabetes is highly effective in reducing the risk of congenital malformations and decreasing the risk of preterm delivery and admission to neonatal intensive care units (NICU). Preconception counseling is also associated with reductions in perinatal mortality and small-for-gestational-age birth weight (3). A key point is the need to incorporate a question about plans for pregnancy into the routine primary and gynecologic care of people with diabetes.

There are opportunities at any health care visit to educate all adults and adolescents with diabetes and childbearing potential about the risks of unplanned pregnancies and about improved maternal and fetal outcomes with pregnancy planning (4). Education and counseling should be offered, even when individuals already use contraception or do not intend to conceive (5). Effective preconception counseling could avert substantial health and associated cost burdens related to the offspring (6). Family planning should be discussed, including the benefits of long-acting, reversible contraception, and effective contraception should be prescribed and used until the individual is prepared and ready to become pregnant (7–12).

All individuals with diabetes and childbearing potential should be informed about the importance of achieving and maintaining as near euglycemia as safely possible prior to conception and throughout pregnancy. Observational studies show an increased risk of diabetic embryopathy, especially anencephaly, microcephaly, congenital heart disease, kidney anomalies, and caudal regression, directly proportional to elevations in A1C during the first 10 weeks of pregnancy (13). Although observational studies are confounded by the association between elevated periconceptional A1C and other engagement in self-care behaviors, the quantity and consistency of data are convincing and support the recommendation to optimize glycemia prior to conception with an A1C <6.5% (<48 mmol/mol), as this is associated with the lowest risk of congenital anomalies (given that organogenesis occurs primarily at 5–8 weeks of gestation), preeclampsia, and preterm birth (13–17). In a systematic review and meta-analysis of observational studies, preconception care for pregnant individuals with preexisting diabetes was associated with lower A1C and reduced risks of birth defects, preterm delivery, perinatal mortality, small-for-gestational-age births, and neonatal intensive care unit admissions (18).

To minimize the occurrence of complications, beginning at the onset of puberty or at diagnosis, all adults and adolescents with diabetes of childbearing potential should receive education about *1*) the risks of malformations associated with unplanned pregnancies, even with mild hyperglycemia, and *2*) the use of effective contraception at all times when trying to prevent a pregnancy. Preconception counseling using developmentally appropriate educational tools enables adolescents with childbearing potential to make well-informed decisions (4). Preconception counseling resources tailored to adolescents are available at no cost through the American Diabetes Association (ADA) (19).

Individuals with prediabetes or a history of GDM will need preconception evaluation for as long as they have childbearing potential. Individuals with a history of GDM who are planning pregnancy should undergo screening for type 2 diabetes or prediabetes prior to conception, and if not screened preconception, testing should be performed before 15 weeks of gestation, as outlined in section 2, "Diagnosis and Classification of Diabetes." In the nonpregnant state, evaluation may be performed with any glycemic test recommended in section 2, "Diagnosis and Classification of Diabetes." Should type 2 diabetes be diagnosed preconception, the individual should initiate treatment with a goal to achieve and maintain an A1C of <6.5% (<48 mmol/mol) prior to conception using therapies approved for use in pregnancy.

Risks for GDM are characterized by an increased risk of large-for-gestational-age birth weight and neonatal and pregnancy complications and an increased risk of long-term maternal type 2 diabetes and abnormal glucose metabolism of offspring in childhood (20). These associations with maternal oral glucose tolerance test (OGTT) results are continuous with no clear inflection points (21,22). Offspring with exposure to untreated GDM have reduced insulin sensitivity and β-cell compensation and are more likely to have impaired glucose tolerance in childhood (23). In other words, short-term and long-term risks increase with progressive maternal hyperglycemia.

Weight should be assessed at the preconception evaluation. Counseling on the specific risks of obesity in pregnancy and lifestyle interventions to prevent and treat obesity, including referral to a registered dietitian nutritionist (RDN), is recommended regardless of diabetes status (24). In a randomized trial of individuals with overweight or obesity and a history of GDM, weight loss prior to a subsequent pregnancy was associated with a lower risk of GDM recurrence, especially when weight loss was ≥5% (odds ratio [OR] 0.18, 95% CI 0.04–0.88) (25). Counseling on weight management should include the known benefits and risks of different strategies for achieving and maintaining weight loss. For strategies that include pharmacotherapy, recommendations should be given for when changes in medications should occur prior to pregnancy. The risk for associated hypertension and other comorbidities may be as high or higher with type 2 diabetes as it is with type 1 diabetes, even if diabetes is better managed and of shorter apparent duration, with pregnancy loss appearing to be more prevalent in the third trimester in those with type 2 diabetes compared with the first trimester in those with type 1 diabetes (26,27).

Preconception Care

Recommendations

15.5 Individuals with preexisting diabetes who are planning a pregnancy should ideally begin receiving interprofessional care prior to conception, which includes an endocrinology health care professional, maternal-fetal medicine specialist, registered dietitian nutritionist, and diabetes care and education specialist, when available. **B**

15.6 In addition to focused attention on achieving glycemic goals, **A** standard preconception care should be augmented with extra focus on nutrition, physical activity, diabetes self-care education, and screening for diabetes comorbidities and complications. **B**

15.7 Individuals with preexisting diabetes who are planning a pregnancy or who have become pregnant should be counseled on the risk of development

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and/or progression of diabetic retinopathy. Dilated eye examinations should occur ideally before pregnancy as well as in the first trimester, and then pregnant individuals should be monitored every trimester and for 1 year postpartum as indicated by the degree of retinopathy and as recommended by the eye care health care professional. **B**

The importance of preconception care for all pregnant people is highlighted by American College of Obstetricians and Gynecologists (ACOG) Committee Opinion 762, "Prepregnancy Counseling" (5). Preconception care for people with prediabetes and diabetes should include the standard screening and care recommended for any person planning pregnancy (5). Prescription of prenatal vitamins with at least 400–800 μg of folic acid (28) and 150 μg of potassium iodide (29) is recommended prior to conception. Review and counseling on abstaining from nicotine products, alcohol, and recreational drugs, including cannabis, is important. Standard care includes screening for sexually transmitted infections and thyroid disease, recommended vaccinations, routine genetic screening, a careful review of all prescription and nonprescription medications, herbal supplements, and nonherbal supplements used and a review of travel history and plans with special attention on areas known to have relevant endemic viruses, as outlined by ACOG. See **Table 15.1** for additional details on elements of preconception care (5,28,30).

Due to the complexity of insulin management in pregnancy, referral to a specialized center offering team-based care (with team members including a maternal-fetal medicine specialist, endocrinologist or other health care professional experienced in managing pregnancy and preexisting diabetes, RDN, diabetes care and education specialist, and social worker, as needed) is recommended if this resource is available. When a single specialized center is not available, providing an interprofessional team approach through interprofessional team members at different centers may still be beneficial.

The most important diabetes-specific component of preconception care is the attainment of glycemic goals prior to conception. Diabetes-specific counseling should include an explanation of the risks to mother and fetus related to pregnancies

Table 15.1—Checklist for preconception care for people with prediabetes, diabetes, or a history of gestational diabetes mellitus

Preconception education should include:
- ☐ Comprehensive nutrition assessment and recommendations for:
  - Overweight and obesity or underweight
  - Meal planning
  - Correction of nutritional deficiencies
  - Caffeine intake
  - Safe food preparation technique
- ☐ Lifestyle recommendations for:
  - Regular moderate exercise
  - Avoidance of hyperthermia (hot tubs)
  - Adequate sleep
- ☐ Comprehensive diabetes self-management education
- ☐ Counseling on diabetes in pregnancy per current standards, including natural history of insulin resistance in pregnancy and postpartum; preconception glycemic goals; avoidance of DKA or severe hyperglycemia; avoidance of severe hypoglycemia; progression of retinopathy in individuals with preexisting diabetes; PCOS (if applicable); fertility in people with diabetes; genetics of diabetes; risks to pregnancy including miscarriage, stillbirth, congenital malformations, macrosomia, preterm labor and delivery, hypertensive disorders in pregnancy
- ☐ Supplementation
  - Folic acid supplement (400–800 μg/day routine)
  - Appropriate use of over-the-counter medications and supplements

Health assessment and plan should include:
- ☐ General evaluation of overall health
- ☐ Evaluation of diabetes and its comorbidities and complications, including DKA or severe hyperglycemia; severe hypoglycemia/hypoglycemia unawareness; barriers to care; comorbidities such as hyperlipidemia, hypertension, MASLD, PCOS, and thyroid dysfunction; complications such as macrovascular disease in individuals with preexisting diabetes, nephropathy, neuropathy (including autonomic bowel and bladder dysfunction), and retinopathy
- ☐ Evaluation of obstetric or gynecologic history, including a history of cesarean section, congenital malformations or fetal loss, current methods of contraception, hypertensive disorders of pregnancy, postpartum hemorrhage, preterm delivery, previous macrosomia, Rh incompatibility, and thrombotic events (DVT/PE)
- ☐ Review of current medications and appropriateness during pregnancy

Screening should include:
- ☐ Diabetes complications and comorbidities, including comprehensive foot exam; comprehensive ophthalmologic exam; ECG in individuals starting at age 35 years who have cardiac signs or symptoms or risk factors and, if abnormal, further evaluation; lipid panel; serum creatinine; TSH; and urine albumin-to-creatinine ratio
- ☐ Anemia
- ☐ Genetic carrier status (based on history):
  - Cystic fibrosis
  - Sickle cell anemia
  - Tay-Sachs disease
  - Thalassemia
  - Others if indicated
- ☐ Infectious disease (per ACOG guidelines)

Preconception plan should include:
- ☐ Immunizations (per ACOG guidelines) (203–205)
- ☐ Nutrition and medication plan to achieve glycemic goals prior to conception, including appropriate implementation of blood glucose monitoring, continuous glucose monitoring (if indicated and appropriate), and pump technology (if indicated and appropriate)
- ☐ Contraceptive plan to prevent pregnancy until glycemic goals are achieved
- ☐ Management plan for general health, gynecologic concerns, comorbid conditions, or complications, if present, including hypertension, nephropathy, retinopathy; Rh incompatibility; and thyroid dysfunction

Created using information from American College of Obstetricians and Gynecologists (ACOG) (5) and others (28,30). DKA, diabetic ketoacidosis; DVT/PE, deep vein thrombosis/pulmonary embolism; ECG, electrocardiogram; MASLD, metabolic dysfunction-associated steatotic liver disease; PCOS, polycystic ovary syndrome; TSH, thyroid-stimulating hormone.

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associated with diabetes and the ways to reduce risks, including glycemic goal setting, lifestyle and behavioral management, and medical nutrition therapy (3). Prior to conception, it is recommended to achieve glucose levels as close to normal as is safely possible without excessive hypoglycemia, ideally A1C <6.5% (<48 mmol/mol), and preconception glycemic goals should be achieved using therapies approved for use in pregnancy. Setting preconception glucose goals is helpful in guiding insulin dose adjustments. The following preconception glucose goals have been advised: preprandial glucose 80–110 mg/dL (4.4–6.1 mmol/L) and 2 h postprandial glucose <155 mg/dL (<8.6 mmol/L) (31). Prandial insulin dose should be reduced for postprandial glucose <100 mg/dL (5.6 mmol/L).

For individuals with preexisting diabetes, the presence of microvascular complications is associated with higher risk of disease progression and adverse pregnancy outcomes (32). Diabetes-specific testing should include A1C, creatinine, and urinary albumin-to-creatinine ratio. Special attention should be paid to the review of the medication list for potentially harmful drugs, e.g., ACE inhibitors (33), angiotensin receptor blockers (33), and lipid-lowering medications including statins in some cases (34). For individuals using medications that are not approved for use in pregnancy (such as some glucose-lowering, lipid-lowering, and antihypertensive agents), preconception care should include recommendations for when changes in medications should occur to stabilize the conditions and risk factors managed by these medications (such as glucose levels, weight, lipids, and blood pressure) on alternate therapies prior to pregnancy. Glucagon-like peptide 1 receptor agonists (GLP-1 RAs) and dual glucose-dependent insulinotropic polypeptide (GIP) and GLP-1 RAs should be discontinued prior to pregnancy, and contraception should be used while taking them (see section 9, "Pharmacologic Approaches to Glycemic Treatment"). The manufacturer prescribing information recommends discontinuation of semaglutide at least 2 months before a planned pregnancy due to its long half-life (35). No recommendation is included in the U.S. prescribing information for tirzepatide, although the Canadian manufacturer prescribing information recommends tirzepatide discontinuation at least 1 month before pregnancy (36). Importantly, before attempting conception, preconception glycemic goals should be achieved following discontinuation of GLP-1 RA therapy or dual GIP and GLP-1 RA therapy, with initiation and titration of insulin therapy when needed to achieve preconception glycemic goals. Thus, individuals planning pregnancy should be counseled that a period of several months is usually needed to allow for both elimination of the medication and adjustment of therapy approved for use in pregnancy to achieve preconception glycemic goals prior to pregnancy. Individuals should additionally be counseled about the risk of weight recurrence after discontinuation of these agents.

A referral for a comprehensive eye exam is recommended. Individuals with preexisting diabetic retinopathy need close monitoring during pregnancy to assess for stability or progression of retinopathy and provide treatment if indicated (37).

GLYCEMIC GOALS IN PREGNANCY

Recommendations

15.8 Fasting, preprandial, and postprandial blood glucose monitoring are recommended in individuals with diabetes in pregnancy to achieve optimal glucose levels. Glucose goals are fasting plasma glucose <95 mg/dL (<5.3 mmol/L) and either 1-h postprandial glucose <140 mg/dL (<7.8 mmol/L) or 2-h postprandial glucose <120 mg/dL (<6.7 mmol/L). **B**

15.9 Due to increased red blood cell turnover, A1C is slightly lower during pregnancy in people with and without diabetes. Ideally, the A1C goal in pregnancy is <6% (<42 mmol/mol) if this can be achieved without significant hypoglycemia, but the goal may be relaxed to <7% (<53 mmol/mol) if necessary to prevent hypoglycemia. **B**

15.10 Continuous glucose monitoring (CGM) can help to achieve glycemic goals (e.g., time in range, time above range) **A** and A1C goal **B** in type 1 diabetes and pregnancy and may be beneficial for other types of diabetes in pregnancy. **E**

15.11 Recommend CGM to pregnant individuals with type 1 diabetes. **A** In conjunction with aims to achieve traditional pre- and postprandial glycemic goals, CGM can reduce the risk for large-for-gestational-age infants and neonatal hypoglycemia in pregnancy complicated by type 1 diabetes. **A**

15.12 CGM metrics may be used in combination with blood glucose monitoring to achieve optimal pre- and postprandial glycemic goals. **E**

15.13 Commonly used estimated A1C and glucose management indicator calculations should not be used in pregnancy as estimates of A1C. **C**

Insulin Physiology

Pregnancy in people with normal glucose metabolism is characterized by fasting glucose levels that are lower than those in the nonpregnant state due to insulin-independent glucose uptake by the fetus and placenta and by mild postprandial hyperglycemia and carbohydrate intolerance as a result of diabetogenic placental factors. Early pregnancy may be a time of enhanced insulin sensitivity and lower glucose levels and is followed by progressive insulin resistance in the second and third trimesters (38–40). Insulin resistance drops rapidly with the delivery of the placenta. In people with normal pancreatic function, insulin production is sufficient to meet the challenge of this physiological insulin resistance and to maintain normal glucose levels. However, in people with diabetes, hyperglycemia occurs if treatment is not adjusted appropriately.

Glucose Monitoring

Reflecting this physiology, fasting and postprandial blood glucose monitoring (BGM) is recommended to achieve glycemic goals in pregnant people with diabetes. Preprandial testing is also recommended when using insulin pumps or basal-bolus therapy so that the premeal rapid-acting insulin dosage can be adjusted. Postprandial monitoring is associated with better glycemic outcomes and a lower risk of preeclampsia (41,42). There are no adequately powered randomized trials comparing different fasting and postmeal glycemic goals for preexisting diabetes in pregnancy.

Similar to the glycemic goals recommended by ACOG (11), the ADA-recommended goals for pregnant people with type 1 or type 2 diabetes are shown in **Table 15.2**. Lower limits are based on the mean of normal blood glucose in pregnancy (43) but do not apply to individuals with type 2 diabetes treated with nutrition alone. Hypoglycemia in pregnancy is as defined and discussed in Recommendations 6.10–6.19 (see section 6, "Glycemic

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Table 15.2—Blood glucose goals in pregnancies associated with diabetes

	Blood glucose goal		
Glucose measurement	Type 1 diabetes or type 2 diabetes^	GDM treated with insulin	GDM not treated with insulin
Fasting glucose	70–95 mg/dL (3.9–5.3 mmol/L)	70–95 mg/dL (3.9–5.3 mmol/L)	<95 mg/dL (<5.3 mmol/L)
1-h postprandial glucose	110–140 mg/dL* (6.1–7.8 mmol/L)	110–140 mg/dL* (6.1–7.8 mmol/L)	<140 mg/dL* (<7.8 mmol/L)
2-h postprandial glucose	100–120 mg/dL (5.6–6.7 mmol/L)	100–120 mg/dL (5.6–6.7 mmol/L)	<120 mg/dL (<6.7 mmol/L)

Gestational diabetes mellitus (GDM) blood glucose goals shown are recommended by the Fifth International Workshop-Conference on Gestational Diabetes Mellitus (45). ^Lower glucose limits do not apply to individuals with type 2 diabetes treated with nutrition alone. Aim for less stringent goals if these cannot be achieved without significant hypoglycemia, based on clinical experience and individualization of care. *Optimal goal includes either a 1-h postprandial glucose level or 2-h postprandial glucose level within column of type of diabetes.

Goals, Hypoglycemia, and Hyperglycemic Crises"). The most appropriate hypoglycemia threshold level in pregnancy has not been validated but has ranged from <60 to <70 mg/dL (<3.3 to <3.9 mmol/L) in the past. Current recommendations for hypoglycemia thresholds include blood glucose <70 mg/dL (<3.9 mmol/L) and sensor glucose <63 mg/dL (<3.5 mmol/L) (43,44). These fasting or premeal and postprandial glucose values represent optimal levels if they can be achieved safely. In practice, it may be challenging for a person with type 1 diabetes to achieve these goals without hypoglycemia, particularly those with a history of recurrent hypoglycemia or impaired awareness of hypoglycemia. If an individual cannot achieve these goals without significant hypoglycemia, aim for less stringent goals based on clinical experience and individualization of care.

For individuals with GDM, the glucose goals recommended by the Fifth International Workshop-Conference on Gestational Diabetes Mellitus should be used (45) (**Table 15.2**).

For individuals found to have hyperglycemia less than overt diabetes on early pregnancy screening, the benefits of treatment remain uncertain, controversy exists regarding optimal management strategies, and professional society guidance is conflicting (46,47). While hyperglycemia less than overt diabetes in early pregnancy has been associated with increased risk of adverse pregnancy and neonatal outcomes and higher risk of a later GDM diagnosis (see Recommendation 2.32b) (48–52), it has not been established that universal treatment in early pregnancy (prior to the usual time of GDM diagnosis and treatment at 24–28 weeks' gestation) is beneficial for all individuals in this group.

Most randomized controlled trials (RCTs) of treatment of early abnormal glucose metabolism have been underpowered for outcomes. One large multicenter RCT (Treatment of Booking Gestational Diabetes Mellitus [TOBOGM] trial), which enrolled pregnant individuals with at least one risk factor for hyperglycemia, administered early screening (mean 15.6 ± 2.5 weeks) for GDM with a 75-g OGTT (53). Individuals who met World Health Organization criteria for GDM were randomized to receive early treatment or a repeat OGTT at 24–28 weeks (with deferred treatment if indicated). The first primary outcome measure was an adverse neonatal composite outcome including birth <37 weeks, birth weight ≥4.5 kg, birth trauma, neonatal respiratory distress within 24 h of birth, phototherapy, stillbirth neonatal death, or shoulder dystocia. Early GDM treatment resulted in a modest improvement in the composite adverse neonatal outcome (24.9% early treatment vs. 30.5% control group, relative risk 0.82 [0.68–0.98]), although this was driven primarily by differences in rates of neonatal respiratory distress between groups that included neonates requiring ≥4 h of supplemental oxygen who may not have required a higher level of respiratory care. In prespecified subgroup analyses, there was a suggestion of more benefit among individuals who had the OGTT at <14 weeks and among those with OGTT glycemic values in higher glycemic ranges (53).

Subsequent secondary analyses have shed additional light on which subgroups may derive greatest benefit from treatment. Importantly, almost one-third of individuals randomized to observation demonstrated regression of GDM when retested at 24–28 weeks' gestation. Pregnancy outcomes among participants with regression were similar to those of individuals without GDM (54), and these individuals are therefore unlikely to benefit from treatment in early gestation. Continuous positive associations were observed between 1- and 2-h postload glucose measurements and the perinatal composite outcome as well as large for gestational age (55). Among those who were randomized to observation, treatment initiation at 24–28 weeks' gestation normalized large-for-gestational-age risk but not risk of preterm birth and neonatal jaundice (56). Systematic reviews and meta-analyses of RCTs have not shown consistent benefits of early GDM treatment (57,58). Taken together with the observation of possible increased risk of small-for-gestational-age infants among TOBOGM participants with OGTT results in the lower glycemic range who were randomized to treatment (53), it appears that treatment benefits are not universal across this population, with suggestion of benefit in some groups and harm in others.

Professional society guidance regarding early pregnancy treatment of hyperglycemia less than overt diabetes sharply differs (46,47). Future studies are needed to refine identification of those individuals with early abnormal glucose metabolism in whom hyperglycemia will persist and in whom early treatment will afford incremental benefit beyond that afforded by treatment for GDM diagnosed at the usual time in late pregnancy. Importantly, the TOBOGM investigators excluded individuals with fasting glucose ≥110 mg/dL (≥6.1 mmol/L) for safety reasons (53), and thus the risk-benefit balance in pregnant individuals with fasting glucose 110–125 mg/dL (6.1–6.9 mmol/L) was not evaluated in this study. While RCT data to guide clinical practice in this subgroup are lacking, it is possible that selective treatment of early pregnancy hyperglycemia within this subgroup would lead to more pronounced reductions in adverse pregnancy and neonatal outcomes.

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Therefore, the benefits of treatment of early abnormal glucose metabolism remain uncertain. Nutrition counseling and periodic testing of fasting glucose levels (e.g., 3–4 times per week) are suggested. Testing frequency may proceed to daily, and treatment may be intensified, if fasting plasma glucose is predominantly ≥110 mg/dL (≥6.1 mmol/L) prior to 15 weeks of gestation. Individualized discussion of areas of uncertainty and the option to treat for fasting glucose ≥110 mg/dL (≥6.1 mmol/L) via shared decision-making is reasonable. When treatment for early abnormal glucose metabolism is initiated, fasting and postprandial glucose levels should be monitored and the standard glucose goals recommended in pregnancies associated with diabetes (**Table 15.2**) should be used.

A1C in Pregnancy

In studies of individuals without preexisting diabetes, increasing A1C levels within the normal range are associated with adverse outcomes (59). In the Hyperglycemia and Adverse Pregnancy Outcome (HAPO) study, increasing levels of glycemia were also associated with worsening outcomes (21). Observational studies in preexisting diabetes and pregnancy show the lowest rates of adverse fetal outcomes in association with A1C <6–6.5% (<42–48 mmol/mol) early in gestation (14,15,17, 60). Clinical trials have not evaluated the risks and benefits of achieving these goals, and treatment goals should account for the risk of maternal hypoglycemia in setting an individualized goal of <6% (<42 mmol/mol) to <7% (<53 mmol/mol). Due to physiological increases in red blood cell turnover, A1C levels fall during normal pregnancy (61,62). Additionally, as A1C represents an integrated measure of glucose, it may not fully capture postprandial hyperglycemia, which drives macrosomia. Thus, although A1C may be useful, it should be used as a secondary measure of glycemia in pregnancy, after glucose monitoring.

In the second and third trimesters, A1C <6% (<42 mmol/mol) has the lowest risk of large-for-gestational-age infants (60,63,64), preterm delivery (65), and preeclampsia (1,66). Taking all of this into account, a goal of <6% (<42 mmol/mol) is optimal during pregnancy if it can be achieved without significant hypoglycemia, which, in addition to the usual adverse sequelae, may increase the risk of low birth weight (67,68). Given the alteration in red blood cell kinetics during pregnancy and physiological changes in glycemic parameters, A1C levels may need to be monitored more frequently than usual (e.g., monthly).

Continuous Glucose Monitoring in Pregnancy

The Continuous Glucose Monitoring in Pregnant Women With Type 1 Diabetes Trial (CONCEPTT) was an RCT of real-time CGM in addition to standard care, including optimization of pre- and postprandial glucose goals, versus standard care for pregnant people with type 1 diabetes. It demonstrated the value of using real-time CGM in pregnant individuals with type 1 diabetes by showing a mild improvement in A1C and significant improvements in the maternal glucose time in range (TIR) and time above range (TAR), without an increase in hypoglycemia, and reductions in large-for-gestational-age births, length of infant hospital stays, and severe neonatal hypoglycemia (69). An observational cohort study that evaluated the glycemic variables reported using CGM systems found that lower mean glucose, lower SD, and higher percentage of TIR were associated with lower risks of large-for-gestational-age births and other adverse neonatal outcomes (70). Data from one study suggest that use of the CGM-reported mean glucose is superior to use of estimated A1C, glucose management indicator, and other calculations to estimate A1C, given the changes to A1C that occur in pregnancy (71). One RCT and several observational studies have found that a 5% increase in CGM TIR was associated with improvements in neonatal morbidity, including large-for-gestational-age births and neonatal intensive care unit admissions (69,70, 72,73). CGM TIR can be used for assessment of glycemic outcomes in people with type 1 diabetes, but it does not provide actionable data to address fasting and postprandial hypoglycemia or hyperglycemia. The cost of CGM use by pregnant individuals with type 1 diabetes is offset by improved maternal and neonatal outcomes (74).

At this time, there are insufficient data to support the use of CGM in all people with type 2 diabetes or GDM (75–79). One RCT evaluating use of CGM or standard BGM in GDM found no significant difference in the primary end point, a composite of adverse perinatal and neonatal outcomes (79). TIR was lower in the CGM group, though CGM use was preferred by participants. In contrast, another RCT of CGM plus adjunctive BGM versus BGM alone in GDM found that 24-h TIR was higher and 24-h mean glucose was lower with CGM use (80). Adequately powered RCTs exclusively evaluating CGM use in pregnant individuals with type 2 diabetes are not currently available. Additionally, it is important to note that optimal targets for CGM metrics during pregnancy affected by type 2 diabetes or GDM have not been defined to date, which confounds interpretation of the limited data available evaluating CGM use in these populations. The decision of whether to use CGM in pregnant individuals with type 2 diabetes or GDM should be individualized based on treatment plan, circumstances, preferences, and needs.

The international consensus on TIR (44) endorses pregnancy glucose goal ranges and goals for TIR for people with type 1 diabetes using CGM as reported on the ambulatory glucose profile. The international consensus on TIR (44) endorses the same sensor glucose goal ranges for individuals with type 2 diabetes in pregnancy and GDM but could not quantify the goal of amount of time spent within each category because of insufficient data. However, the consensus does not specify the type or accuracy of the CGM device or need for alarms and alerts. The accuracy of one CGM system among pregnant individuals with type 1 diabetes, type 2 diabetes, and GDM has been demonstrated, supporting its use in pregnancy without confirmatory capillary blood glucose measurements (81). The proportion of CGM values within 20% or 20 mg/dL of comparator values ≥100 mg/dL or <100 mg/dL, respectively (%20/20 agreement rates), was 78.6% on day 1 of sensor wear, increased to 96.3% on days 4 and 7, and increased to 97.3% on day 10. Although CGM systems for use in pregnancy do not require calibrations and are approved for nonadjunctive use, determination of glucose levels by finger stick is advised in certain circumstances, such as following correction of hypoglycemia (due to the time lag between change in plasma vs. interstitial glucose values), when symptoms do not match sensor glucose reading, and on the first day of use when sensor accuracy may be lower (81). Selection of CGM device should be based on an individual's circumstances, preferences, and needs.

Goals for sensor glucose ranges in pregnancy are:

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- Goal sensor glucose range 63–140 mg/dL (3.5–7.8 mmol/L): TIR, goal >70%
- Time below range (TBR) (<63 mg/dL [<3.5 mmol/L]): level 1 TBR, goal <4%
- TBR (<54 mg/dL [<3.0 mmol/L]): level 2 TBR, goal <1%
- TAR (>140 mg/dL [>7.8 mmol/L]): TAR, goal <25%

Goals for time spent in each range are specific for pregnant individuals with type 1 diabetes.

MANAGEMENT OF DIABETES IN PREGNANCY

Recommendations

15.14 Nutrition counseling before and during pregnancy should promote an eating pattern including fruits, vegetables, legumes, whole grains, nuts, seeds, fish, and other lean protein, which will provide a balance of macronutrients and healthy n-3 fatty acids. **C**

15.15 Lifestyle behavior change is an essential component of management of GDM and may suffice as treatment for many individuals. Insulin should be added if needed to achieve glycemic goals. **A**

15.16 Telehealth visits used in combination with in-person visits for pregnant people with GDM can improve outcomes compared with standard in person care alone. **A**

15.17 Insulin should be used to manage type 1 diabetes in pregnancy **A** and is the preferred agent for the management of GDM **A** and type 2 diabetes in pregnancy. **B**

15.18 Either multiple daily injections or insulin pump technology can be used in pregnancy complicated by type 1 diabetes. **C**

15.19 Automated insulin delivery (AID) systems with pregnancy-specific glucose targets are recommended for pregnant individuals with type 1 diabetes. **A**

15.20 AID systems without pregnancy-specific glucose targets or a pregnancy-specific algorithm may be considered for select pregnant individuals with type 1 diabetes when used with assistive techniques and working with experienced health care teams. **B**

15.21 Metformin and glyburide, individually or in combination, should not be used as first-line agents for management of diabetes in pregnancy, as both cross the placenta to the fetus **A** and may not be sufficient to achieve glycemic goals. **B** Other oral and noninsulin injectable glucose-lowering medications lack long-term safety data and are not recommended. **E**

15.22 Metformin, when used to treat polycystic ovary syndrome and induce ovulation, should be discontinued by the end of the first trimester. **A**

The management of pregnancies associated with diabetes includes appropriate nutrition, lifestyle and behavior management, physical activity goals, and pharmacotherapy to support the maternal, fetal, and placental needs and reach glycemic goals regardless of the diabetes type.

Medical Nutrition Therapy

In people with preexisting diabetes, glycemic goals are usually achieved through a combination of insulin administration and medical nutrition therapy. Because glycemic goals in pregnant individuals are stricter than in nonpregnant individuals, it is important that pregnant people with diabetes consume consistent amounts of carbohydrates to match their prandial insulin dosages or use insulin-to-carbohydrate ratios to match prandial insulin dosages to the carbohydrates being consumed in order to minimize postprandial hyperglycemia or hypoglycemia. Referral to an RDN is important to establish a food plan and to determine weight gain goals. The quality of the carbohydrates should be evaluated. A subgroup analysis of the CONCEPTT study demonstrated that the food choices of individuals planning pregnancy and currently pregnant assessed during the run-in phase prior to randomization were characterized by high-fat, low-fiber, and poor-quality carbohydrate intakes. Fruit and vegetable consumption was inadequate, with one in four participants at risk for micronutrient deficiencies, highlighting the importance of medical nutrition therapy (82).

An expert panel on nutrition in pregnancy and the U.S. Department of Health and Human Services recommend a balance of macronutrients. An eating pattern that severely restricts any macronutrient class should be avoided, specifically the ketogenic eating pattern that lacks carbohydrates, the paleo eating pattern because of dairy restriction, and any eating pattern characterized by excess saturated fats (83). Pregnant individuals with diabetes are recommended to consume whole foods, including fruits, vegetables, legumes, whole grains, lean protein, and healthy fats with n-3 fatty acids, which includes nuts and seeds and fish, which are less likely to promote excessive weight gain (84). Processed foods, fatty red meat, and sweetened foods and beverages should be limited (83,84).

The recommended dietary reference intake for all pregnant people is a minimum of 175 g of carbohydrate (~35% of a 2,000-calorie diet), a minimum of 71 g of protein, and 28 g of fiber (85). The nutrition plan should emphasize monounsaturated and polyunsaturated fats while limiting saturated fats and avoiding *trans* fats. As is true for all nutrition therapy in people with diabetes, the amount and type of carbohydrate will impact glucose levels. Promoting higher-quality, nutrient-dense carbohydrates results in improved fasting or postprandial glucose levels, lower free fatty acids, improved insulin action, and vascular benefits, and it may reduce excess infant adiposity. Individuals who substitute fat for carbohydrates may unintentionally enhance lipolysis, promote elevated free fatty acids, and worsen maternal insulin resistance (86,87). Fasting ketone testing may be useful to identify those who are severely restricting carbohydrates to manage blood glucose. Carbohydrate restriction can increase the risk of higher dietary fat consumption, which may lead to fetal overgrowth (83). Simple carbohydrates will result in higher postmeal excursions.

Medical nutrition therapy for GDM is an individualized nutrition plan developed between the pregnant person and an RDN familiar with the management of GDM (88, 89). The food plan should provide adequate calorie intake to promote fetal, neonatal, and maternal health, achieve glycemic goals, and promote appropriate weight gain, according to the 2009 National Academy of Medicine recommendations (90). There is no definitive research that identifies a specific optimal calorie intake for people with GDM or suggests that their calorie needs are different from those of pregnant individuals without GDM. The food plan should be based on a nutrition assessment with dietary reference intake guidance from the National Academy of Medicine.

Lifestyle and Behavioral Management

Although there is some heterogeneity, many RCTs and a Cochrane review suggest that the risk of GDM may be reduced by appropriate nutrition, exercise, and

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lifestyle counseling, particularly when interventions are started during the first trimester or early in the second trimester (91–93).

After diagnosis of GDM, treatment starts with medical nutrition therapy, physical activity, and weight management, depending on pregestational weight, as outlined in this section. Depending on the population, studies suggest that 70–85% of people diagnosed with GDM under Carpenter-Coustan criteria can manage GDM with lifestyle modification alone; it is anticipated that this proportion will be even higher if the lower International Association of the Diabetes and Pregnancy Study Groups (94) diagnostic thresholds are used.

Physical Activity

It is recommended that generally healthy people do at least 150 min of moderate-intensity aerobic activity each week during pregnancy and postpartum, preferably spread throughout the week (95). Adjustments to a physical activity routine or plan should be done in consultation with a health care professional, especially if someone is considering a big change in physical activity intensity (95). Such activity improves cardiorespiratory fitness and reduces the risk for excessive gestational weight gain or postpartum weight retention (95).

Regular exercise has been reported to reduce the odds of developing GDM, gestational hypertension, and preeclampsia by 38%, 39%, and 41%, respectively (96), and acute and long-term prenatal exercise can positively affect glycemia, with a larger effect noted in people with diabetes (97). With respect to GDM, a systematic review demonstrated improvements in glucose outcomes and reductions in need to start insulin or insulin dose requirements with an exercise intervention. However, there was heterogeneity in the types of effective exercise (aerobic, resistance, or both) and duration of exercise (20–50 min/day, 2–7 days/week of moderate intensity) (98), so there is insufficient evidence about which specific type of exercise program has the biggest impact on these diabetes-related outcomes in pregnancy.

Health Care Delivery for People With Diabetes in Pregnancy

As discussed in PRECONCEPTION CARE, team-based care is recommended either through a single specialized center (when available) or multiple centers with interprofessional team members as part of the care plan during pregnancy. A meta-analysis of 32 RCTs evaluating the effectiveness of telehealth interventions, which ranged from telehealth visits to the use of health apps, used in combination with in-person visits for GDM, demonstrated reduced incidences of cesarean delivery, premature rupture of membranes, pregnancy-induced hypertension or preeclampsia, preterm birth, neonatal asphyxia, and polyhydramnios compared with standard in-person care alone (99).

Pharmacologic Therapy

Insulin

Insulin should be used to manage type 1 diabetes in pregnancy and is preferred for the management of type 2 diabetes in pregnancy and GDM. The physiology of pregnancy necessitates frequent titration of insulin to match changing requirements and underscores the importance of daily and frequent glucose monitoring. In early pregnancy, many people with type 1 diabetes will have lower insulin requirements and an increased risk for hypoglycemia (38). At around 16 weeks, insulin resistance begins to increase, and total daily insulin doses increase linearly by ∼5% per week through week 36. This usually results in a doubling of daily insulin dose compared with the prepregnancy requirement. While there is an increase in both basal and bolus insulin requirements, bolus insulin requirements take up a larger proportion of overall total daily insulin needs in individuals with preexisting diabetes as pregnancy progresses (39,40). The insulin requirement levels off toward the end of the third trimester. A rapid and significant reduction in insulin requirements may indicate the development of placental insufficiency (100), although data are conflicting (101).

Optimal glycemic goals are often easier to achieve during pregnancy with type 2 diabetes than with type 1 diabetes but can require much higher doses of insulin, sometimes necessitating concentrated insulin formulations. It is recommended that insulin management be performed with interprofessional team members with relevant expertise.

None of the currently available human insulin preparations have been demonstrated to cross the placenta (102–107). Insulins studied in RCTs are preferred (108–110) over those studied in cohort studies (111), which are preferred over those studied in case reports only. Basal insulin options in pregnancy include insulin glargine (long-acting), NPH insulin (intermediate-acting), and insulin degludec (ultralong-acting). Insulin detemir (the only basal insulin with RCT data supporting its use during pregnancy) was removed from the market by the manufacturer and is no longer available. Observational studies of glargine use during pregnancy have not shown evidence of increased adverse fetal outcomes with use of insulin glargine versus NPH insulin (111), and insulin glargine is widely used during pregnancy. Although the data on insulin degludec are still limited and emerging, insulin degludec was shown in an RCT to be noninferior to detemir (108), noting that this study was not powered to detect differences in maternal safety or pregnancy outcomes. The prolonged duration of action of insulin degludec (at least 42 h) may present challenges in pregnancy. The need for slower dose titration due to its longer half-life (recommended interval between dose changes is 3–4 days [112]) may impede the frequent dose adjustments often needed in response to the rapidly changing insulin requirements of pregnancy, and hypoglycemia could occur postpartum when insulin requirements decrease sharply. Regarding rapid-acting insulins options, insulin aspart and insulin lispro are widely considered safe and effective for use during pregnancy (113–115). Use of faster-acting insulin aspart (faster aspart) among pregnant women with type 1 or type 2 diabetes led to similar fetal growth and A1C relative to insulin aspart (116); a secondary analysis of women with type 1 diabetes showed that faster-acting aspart use resulted in higher TIR and less severe hypoglycemia compared with aspart (117).

Both multiple daily insulin injections and continuous subcutaneous insulin infusion are reasonable delivery strategies in pregnancy, with neither showing superiority over the other (106,118).

Automated insulin delivery (AID) systems have been studied in pregnancy and postpartum. The use of AID systems in pregnancy affected by diabetes presents particular challenges, as the current U.S. Food and Drug Administration (FDA)-approved AID systems (except for one that has been FDA approved but is not yet commercially available in the U.S.) do not have algorithms designed to achieve pregnancy-specific glucose goals,

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and the lowest glucose target available for most systems is higher than the fasting glucose target range recommended during pregnancy. Initiating or continuing AID systems during pregnancy therefore needs to be assessed carefully. Selected individuals with type 1 diabetes should be evaluated as potential candidates for AID systems in the setting of expert guidance.

Recent data have shown the clinical benefits and safety of AID use, though only one study used an AID system with a pregnancy-specific glucose target. In the Automated insulin Delivery Among Pregnant women with Type 1 diabetes [AiDAPT] trial, 124 pregnant individuals with type 1 diabetes were randomized to either an AID system with glucose targets that could be set near or in the pregnancy-specific fasting glucose range or standard of care (CGM with either multiple daily injections or standard insulin pump therapy) (119). Investigators recommended pump glucose targets of 100 mg/dL in early pregnancy and 81–90 mg/dL from 16 to 20 weeks onward. The AID group had a higher TIR (10.5% difference between groups, $P < 0.001$), lower TAR (−10.2% [95% CI −13.8 to −6.6]), and lower A1C (−0.31% [−0.50 to −0.12]), and a subset of participants who were interviewed reported benefits with AID use during pregnancy (e.g., more enjoyment of pregnancy, better sleep, less worry) (119, 120). The AID system evaluated in this study is FDA approved but is not currently available in the U.S.

There have additionally been RCTs examining AID systems that do not have either pregnancy-specific glucose targets in the algorithms or algorithms that adapt specifically to pregnancy but were used with assistive techniques. In a study with 95 pregnant individuals with type 1 diabetes, participants used an AID system set to a glucose target of 100 mg/dL or to standard of care. The 24-h TIR was similar between groups, but the nocturnal TIR was higher (6.58%, $P = 0.003$), the 24-h TBR was lower (−1.34%, $P = 0.002$), and the nocturnal TBR was lower (−1.86%, $P = 0.0005$) in the AID group (121). AID users reported higher diabetes treatment satisfaction and had less hypoglycemia unawareness (per Gold scores) (121). In a pilot study ($n = 23$) where participants were randomized in the second trimester to AID with a system whose glucose target is 120 mg/dL or sensor-augmented pump (SAP) therapy with the same system, time spent <63 mg/dL (<3.5 mmol/L) decreased significantly in the AID group from baseline to third trimester (7.5% first trimester vs. 2.8% third trimester, $P < 0.05$), but the average sensor glucose was higher in the AID group in the third trimester (mean [SE] 119 [4] SAP vs. 132 [4] AID, $P = 0.0475$) without significant differences between groups in other CGM metrics (122). These two studies used assistive techniques, such as administration of fake carbohydrate insulin boluses for carbohydrates that were not consumed, and pump management was determined by expert guidance from an experienced interprofessional team (121–123). Thus, it may be appropriate to continue or initiate AID therapy with systems that do not have pregnancy-specific glucose targets or algorithms in carefully selected pregnant individuals with type 1 diabetes in the setting of using assistive techniques with expert guidance (121–123). Some pregnant individuals may derive glycemic benefit from use of an AID system that is not approved for use in pregnancy (for example, decreased hypoglycemia and/or improved TIR in automated vs. manual mode in that individual); decisions regarding AID use during pregnancy should be informed by careful consideration of individualized risks and benefits. Assessment of potential candidates for AID use in pregnancy should include relevant parameters such as glycemic levels, presence or absence of severe hypoglycemic or hyperglycemic events, ability or comfort in engaging with diabetes technology, psychosocial determinants, cost, individual preference, and other factors as relevant. If the decision is made to use AID systems without pregnancy-specific targets in selected pregnant individuals, then using assistive techniques, such as the combination of SAP mode (or manual mode) and automated mode at different time points in pregnancy or throughout the day, should be considered and applied as needed to achieve intended goals (123).

Continuous subcutaneous insulin infusion was compared with intravenous insulin infusion in an RCT of 70 participants during labor and delivery. There was no difference between groups in the primary outcome of neonatal hypoglycemia or in secondary outcomes (e.g., mean neonatal glucose in first 24 h of life, severe neonatal hypoglycemia) (124). In an RCT of 18 participants using AID or SAP therapy for 12 weeks postpartum (125), those in the AID group had fewer hypoglycemia episodes. One prespecified observational analysis of individuals who continued AID intrapartum and/or early postpartum found improved TIR intrapartum and similar TIR early postpartum in the AID group compared with the standard insulin therapy group, without severe hypoglycemia (126). In a prespecified extension to the AiDAPT trial, in which participants continued either AID or standard insulin therapy for 6 months postpartum, TIR was higher and mean glucose lower in the AID group, without significant differences in hypoglycemia (127). Suggested strategies for postpartum adjustment of AID pump settings have been described (128).

Treatment of GDM with lifestyle and insulin has been demonstrated to improve perinatal outcomes in two large RCTs, as summarized in a U.S. Preventive Services Task Force review (129). Insulin is the first-line agent recommended for the treatment of GDM in the U.S. While individual studies support limited efficacy of metformin (130,131) and glyburide (132) in reducing glucose levels for the treatment of GDM, these agents are not recommended as the first-line treatment of GDM because they are known to cross the placenta and data on long-term safety for offspring is of some concern (20). Furthermore, in separate RCTs, glyburide and metformin failed to achieve adequate glycemic outcomes in 23% and 25–28% of participants with GDM, respectively (133,134).

Sulfonylureas

Sulfonylureas are known to cross the placenta and have been associated with increased neonatal hypoglycemia. Concentrations of glyburide in umbilical cord plasma are approximately 50–70% of maternal levels (133,134). In systematic reviews and meta-analyses, compared with insulin or metformin, glyburide was associated with a higher rate of neonatal hypoglycemia and macrosomia and an increased neonatal abdominal circumference (135,136).

Glyburide was not found to be noninferior to insulin based on a composite outcome of neonatal hypoglycemia, macrosomia, and hyperbilirubinemia among individuals with GDM (137). In a randomized noninferiority trial evaluating insulin compared with an oral glucose-lowering medication strategy (metformin with addition of glyburide and then insulin substitution for glyburide if glycemic goals were not met) in GDM, the oral glucose-lowering medication strategy failed to meet criteria for noninferiority compared with insulin in

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preventing large-for-gestational-age infants (138). In addition, the oral glucose-lowering strategy was associated with increased risk of maternal hypoglycemia compared with insulin (138). Long-term safety data for offspring exposed to glyburide are not available (137,138).

Metformin

Metformin was associated with a lower risk of neonatal hypoglycemia and less maternal weight gain than insulin in systematic reviews and RCTs for GDM treatment, but treatment monotherapy failure occurred in 14–46% of individuals (135, 139–142). A meta-analysis of 11 RCTs demonstrated that metformin treatment in pregnancy does not reduce the risk of GDM in high-risk individuals with obesity, polycystic ovary syndrome, or preexisting insulin resistance (143). RCTs of individuals with preexisting type 2 diabetes treated either with insulin alone or insulin plus metformin did not show differences in composite neonatal health outcomes between groups (144,145), and one of these also included individuals diagnosed with diabetes early in gestation (145). Neonatal birth weights were smaller in the metformin groups of these studies, but the metformin group experienced more drug intolerance in one study and there was a doubling of small-for-gestational-age neonates in the other (144,145). RCTs comparing metformin with other therapies for ovulation induction in individuals with polycystic ovary syndrome have not demonstrated benefit in preventing spontaneous abortion or GDM (146), and there is no evidence-based need to continue metformin in these individuals (147–149).

Of note, metformin readily crosses the placenta, resulting in umbilical cord blood levels of metformin as high or higher than simultaneous maternal levels (150,151). In a primate model, metformin initiated early in gestation led to fetal bioaccumulation of metformin, growth restriction, and renal dysmorphology (152). In the Metformin in Gestational Diabetes: The Offspring Follow-Up (MiG TOFU) study's analyses of 7- to 9-year-old offspring, the 9-year-old offspring exposed to metformin for the treatment of GDM in the Auckland cohort (but not the Adelaide cohort) were heavier and had a higher waist-to-height ratio and waist circumference than those exposed to insulin (153). In one RCT of metformin use in pregnancy for polycystic ovary syndrome, follow-up of 4-year-old offspring demonstrated higher BMI and increased obesity in the offspring exposed to metformin (154). A follow-up study at 5–10 years showed that the offspring had higher BMI, weight-to-height ratios, and waist circumferences and a borderline increase in fat mass (155,156). A meta-analysis demonstrated that metformin exposure resulted in smaller neonates with an acceleration of postnatal growth, resulting in higher BMI in childhood (155). Follow-up of offspring from the Metformin in Women with Type 2 Diabetes in Pregnancy (MiTy Kids) trial showed no differences in anthropometrics of children at 24 months (157). When prepregnancy metformin use is continued in early pregnancy, it can be discontinued at the end of the first trimester (after organogenesis) to avoid sudden deterioration of glycemia and an associated increased risk of malformations due to hyperglycemia (158). Continuation of metformin in early pregnancy is considered safe, as first-trimester metformin use has not been associated with increased risk of major congenital malformations (159,160), and there is minimal maternal-to-fetal transfer of metformin in early gestation (161,162).

There are some people with GDM requiring medical therapy who may not be able to use insulin safely or effectively during pregnancy due to cost, comprehension, or cultural influences. Oral agents may be an alternative for these individuals after discussing the known risks and the need for more long-term safety data in offspring. However, due to the potential for growth restriction or acidosis in the setting of placental insufficiency, metformin should not be used in pregnant people with hypertension or preeclampsia or those at risk for intrauterine growth restriction (157,161,162).

Special Considerations for Management of Pregnancies With Diabetes

Pregnant individuals with type 1 diabetes have an increased risk of hypoglycemia in the first trimester and after delivery, and like all pregnant people, they have altered counterregulatory response in pregnancy that may decrease hypoglycemia awareness. Education for people with diabetes and family members about the prevention, recognition, and treatment of hypoglycemia is important before, during, and after pregnancy to help prevent and manage hypoglycemia risk.

Pregnancy is a ketogenic state, and people with type 1 diabetes and, to a lesser extent, those with type 2 diabetes are at risk for diabetic ketoacidosis (DKA) at lower blood glucose levels than in the nonpregnant state. Pregnant people with type 1 diabetes should be advised to obtain ketone test strips and receive education on DKA prevention and detection. DKA carries a high risk of stillbirth. Those in DKA who are unable to eat often require intravenous fluid containing dextrose with an insulin drip to adequately meet the higher carbohydrate demands of the placenta and fetus in the third trimester to resolve their ketosis.

Retinopathy is a special concern in pregnancy. The necessary rapid implementation of euglycemia in the setting of retinopathy is associated with worsening of retinopathy (163). Meta-analyses have also demonstrated a high risk of new-onset retinopathy and progression of existing retinopathy in pregnant individuals with type 1 or type 2 diabetes (37,164). Therefore, it is recommended that individuals with preexisting diabetes have dilated eye examinations before pregnancy, in each trimester of pregnancy, and for 1 year postpartum as indicated by the degree of retinopathy and as recommended by the eye care health care professional.

Recommended weight gain during pregnancy for people with overweight status is 15–25 lbs (6.8–11.3 kg) and for those with obesity is 10–20 lbs (4.5–9.1 kg) (90). There are no adequate data on optimal weight gain versus weight maintenance in pregnant people with BMI >35 kg/m^2; however, losing weight is not recommended because of the increased risk of small-for-gestational-age infants (24).

PREECLAMPSIA AND ASPIRIN

Recommendation

15.23 Pregnant individuals with type 1 or type 2 diabetes should be prescribed low-dose aspirin 100–150 mg/day starting at 12–16 weeks of gestation to lower the risk of preeclampsia. **E** A dosage of 162 mg/day may be acceptable; **E** currently, in the U.S., low-dose aspirin is available in 81-mg tablets.

Diabetes in pregnancy is associated with an increased risk of preeclampsia (165). The U.S. Preventive Services Task Force recommends that blood pressure measurements be obtained throughout gestation to screen for hypertensive disorders of pregnancy

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(166). The Task Force also recommends using low-dose aspirin (81 mg/day) as a preventive medication at 12 weeks of gestation in individuals at high risk for preeclampsia, such as those with type 1 or type 2 diabetes (167). However, a meta-analysis and an additional trial demonstrate that low-dose aspirin <100 mg is not effective in reducing preeclampsia, so a dose of >100 mg is required (168–170). A cost-benefit analysis has concluded that this approach would reduce morbidity, save lives, and lower health care costs (171). There are insufficient data about whether the use of aspirin specifically in pregnant people with preexisting diabetes ultimately reduces the incidence of preeclampsia (172,173), although a meta-analysis showed that preeclampsia reductions occurred with aspirin administration in high-risk groups overall (165). Individuals with GDM may be candidates for aspirin therapy for preeclampsia prevention if they have a single high-risk factor, such as chronic hypertension or an autoimmune disease, or multiple moderate risk factors, such as being nulliparous, having obesity, being age ≥35 years, or other factors per the U.S. Preventive Services Task Force (167). More studies are needed to assess the long-term effects of prenatal aspirin exposure on offspring (172).

PREGNANCY AND NON–GLUCOSE-LOWERING DRUG CONSIDERATIONS

Recommendations

15.24 In pregnant individuals with diabetes and chronic hypertension, a blood pressure threshold <140/90 mmHg is recommended for initiation and titration of therapy for better pregnancy outcomes. **A** Therapy should be deintensified if blood pressure is <90/60 mmHg. **E**

15.25a Potentially harmful medications in pregnancy (e.g., ACE inhibitors, angiotensin receptor blockers, mineralocorticoid receptor antagonists) should be stopped prior to conception and avoided in sexually active individuals of childbearing potential who are not using reliable contraception. **B**

15.25b In most circumstances, lipid-lowering medications should be stopped prior to conception and avoided in sexually active individuals of childbearing potential with diabetes who are not using reliable contraception. **B** In some circumstances (familial hypercholesterolemia, severe hypertriglyceridemia, prior atherosclerotic cardiovascular disease event), lipid-lowering therapy may be continued when the benefits outweigh risks. **E**

In normal pregnancy, blood pressure is lower than in the nonpregnant state. The Chronic Hypertension and Pregnancy (CHAP) Trial Consortium's RCT on treatment of mild chronic hypertension during pregnancy demonstrated that a blood pressure of 140/90 mmHg, as the threshold for initiation or titration of treatment, reduces the incidence of adverse pregnancy outcomes without compromising fetal growth (174). The CHAP Consortium's study mitigates concerns about small-for-gestational-age birth weight. Attained mean ± SD blood pressure measurements in the treated versus untreated groups were systolic 129.5 ± 10.0 vs. 132.6 ± 10.1 mmHg (between-group difference −3.11 [95% CI −3.95 to 2.28]) and diastolic 79.1 ± 7.4 vs. 81.5 ± 8.0 mmHg (−2.33 [95% CI −2.97 to 0.04]), respectively (174). Individuals with diabetes had an even better composite outcome score than those without diabetes (174).

A secondary analysis of the CHAP trial evaluated 434 individuals with both diabetes and chronic hypertension to determine the effect of a lower blood pressure goal (<130/80 mmHg) on pregnancy and neonatal outcomes (175). Participants with lower blood pressure were more likely to be on antihypertensive medications at the start of pregnancy and more likely to have newly diagnosed diabetes prior to 20 weeks. Average blood pressure attained was systolic 122.0 ± 5.7 mmHg vs. 134.9 ± 7.4 mmHg and diastolic 74.4 ± 4.0 mmHg vs. 81.7 ± 5.7 mmHg in the <130/80 mmHg vs. 130–139/80–89 mmHg groups, respectively. Participants with an average blood pressure <130/80 mmHg were less likely to have the primary composite adverse perinatal outcome (19.3% vs. 46.5%, adjusted relative risk 0.43, 95% CI 0.30–0.61) and its components: preeclampsia with severe features (14.0 vs. 40.5%, adjusted risk ratio 0.35, 95% CI 0.23–0.54) and indicated preterm birth <35 weeks (8.0% vs. 20.1%, adjusted risk ratio 0.44, 95% CI 0.24–0.79), $P < 0.01$ for both. They were less likely to have adverse perinatal outcomes: birth weight <10th percentile (3.3% vs. 8.8%, adjusted risk ratio 0.37, 95% CI 0.14–0.96, $P = 0.04$), NICU admission (36.0% vs. 51.4%, adjusted risk ratio 0.74, 95% CI 0.59–0.94, $P = 0.01$), and days in NICU (7.0 vs. 11.0, adjusted risk ratio 0.54, 95% CI 0.49–0.59, $P < 0.01$) (175). As participants were not randomized to a blood pressure goal of <130/80 mmHg, there is uncertainty as to whether the improved outcomes were attributable to the lower attained blood pressures or to other underlying differences in participant characteristics. There was no evidence of worse outcomes in the group with blood pressure <130/80 mmHg.

As a result of the CHAP study, ACOG issued a Practice Advisory recommending a blood pressure of 140/90 mmHg as the threshold for initiation or titration of medical therapy for chronic hypertension in pregnancy (176) rather than their previously recommended threshold of 160/110 mmHg (177).

During pregnancy, treatment with ACE inhibitors and angiotensin receptor blockers is contraindicated because they may cause fetal renal dysplasia, oligohydramnios, pulmonary hypoplasia, and intrauterine growth restriction (33). A large study found that after adjusting for confounders, first-trimester ACE inhibitor exposure does not appear to be associated with congenital malformations (178). ACE inhibitors and angiotensin receptor blockers should be stopped prior to pregnancy or as soon as possible in the first trimester to avoid second- and third-trimester fetopathy (178). Antihypertensive drugs known to be effective and safe in pregnancy include methyldopa, nifedipine, labetalol, and clonidine. Atenolol is not recommended, but other β-blockers may be used, if necessary. Diuretic use during pregnancy is generally not recommended, although it may be used safely when prescribed at lower doses for individuals in certain circumstances (e.g., chronic kidney disease and reduced glomerular filtration rate) (179).

For most individuals, lipid-lowering medications (e.g., bempedoic acid, PCSK9 therapies, fibrates) should be stopped prior to pregnancy or at the first pregnancy visit (34). Based on available evidence, statins should also be avoided in pregnancy in most circumstances (34). The risk of teratogenicity with statins appears to be low, but data are limited (34). Statins can be considered in a shared decision-making process between pregnant people with diabetes and their health care teams, including

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discussion of risks and benefits in pregnant individuals at high risk, such as those with a history of atherosclerotic cardiovascular disease or familial hypercholesterolemia (homozygous or severe heterozygous) (34). Hydrophilic statins, such as pravastatin, may be associated with less fetal harm than lipophilic statins (180). Pravastatin has been studied in multiple pregnancy trials administering therapy at various time points in gestation with the aim to reduce preeclampsia risk, and although its ability to do so is inconclusive to date, there does not appear to be increased neonatal mortality or morbidity associated with its use during gestation (34). In pregnant individuals with a history of severe hypertriglyceridemia, triglyceride levels should be monitored, and nutrition interventions with or without pharmacologic therapy for severe hypertriglyceridemia should be instituted when indicated to reduce pancreatitis risk (34).

POSTPARTUM CARE

Recommendations

15.26 Insulin requirements need to be evaluated and adjusted for individuals requiring insulin after delivery because insulin resistance decreases dramatically immediately postpartum. **C**

15.27 A contraceptive plan should be discussed and implemented with all people with diabetes of childbearing potential. **A**

15.28 Breastfeeding efforts are recommended for all individuals with diabetes. **A** Breastfeeding is recommended for individuals with a history of GDM for multiple benefits, **A** including a reduced risk for type 2 diabetes later in life. **B**

15.29 Postpartum care should include psychosocial assessment and support for self-care. **E**

15.30 Screen individuals with a recent history of GDM at 4–12 weeks postpartum, using the 75-g oral glucose tolerance test and clinically appropriate nonpregnancy diagnostic criteria. **B**

15.31 Individuals with a history of GDM should have lifelong screening for the development of type 2 diabetes or prediabetes every 1–3 years. **B**

15.32 Individuals with overweight or obesity and a history of GDM found to have prediabetes should receive intensive lifestyle interventions and/or metformin to prevent diabetes. **A**

Diabetes Treatment Postpartum

For individuals requiring insulin after delivery, insulin sensitivity increases dramatically with the delivery of the placenta. In one study, insulin requirements in the immediate postpartum period were roughly 34% lower than prepregnancy insulin requirements (181). Insulin sensitivity then returns to prepregnancy levels over the following 1–2 weeks. For individuals taking insulin, particular attention should be directed to hypoglycemia prevention in the setting of breastfeeding and erratic sleep and eating schedules (182). Individuals with GDM usually do not require diabetes medications in the postpartum period.

Contraception

A major barrier to effective preconception care is the fact that many pregnancies are unplanned (183). Planning pregnancy is critical in individuals with preexisting diabetes to achieve the optimal glycemic goals necessary to prevent congenital malformations and reduce the risk of other complications. Therefore, all individuals with diabetes of childbearing potential should have family planning options reviewed at regular intervals to make sure that effective contraception is implemented and maintained. This applies to individuals in the immediate postpartum period. Individuals with diabetes have the same contraception options and recommendations as those without diabetes, although the existence of diabetes complications or other vascular disease may modify recommended options (12). Long-acting, reversible contraception may be ideal for individuals with diabetes and childbearing potential. The risk of an unplanned pregnancy outweighs the risk of any currently available contraception option.

Lactation

Considering the immediate nutritional and immunological benefits of breastfeeding for the baby, all mothers, including those with diabetes, should be supported in attempts to breastfeed. An analysis of 28 systematic reviews and meta-analyses of associations between breastfeeding and outcomes in children found that breastfeeding was associated with numerous health benefits for children, such as reduced infant mortality due to infectious diseases at <6 months of age (OR 0.22–0.59 across studies), reduced respiratory infections in children aged <2 years, and reduced asthma or wheezing in children and adolescents aged 5–18 years (OR 0.91, 0.85–0.98) (184). The same analysis found that breastfeeding was associated with improved maternal health outcomes, including reduced risks of breast cancer (OR 0.81 [95% CI 0.77–0.86]), ovarian cancer (OR 0.70 [95% CI 0.64–0.75]), and type 2 diabetes (OR 0.68 [95% CI 0.57–0.82]). Breastfeeding may also confer longer-term metabolic benefits to both mother (185) and offspring (186). Breastfeeding reduces the risk of developing type 2 diabetes in mothers with previous GDM (185). It may improve the metabolic risk factors of offspring, but more studies are needed (187). However, lactation can increase the risk of overnight hypoglycemia, and insulin dosing may need to be adjusted.

Special Postpartum Considerations for Individuals With Gestational Diabetes Mellitus

Because GDM often represents previously undiagnosed prediabetes, type 2 diabetes, maturity-onset diabetes of the young, or even developing type 1 diabetes, individuals with GDM should be tested for persistent diabetes or prediabetes at 4–12 weeks postpartum with a fasting 75-g OGTT using nonpregnancy criteria as outlined in section 2, "Diagnosis and Classification of Diabetes," specifically **Tables 2.1** and **2.2**. The OGTT is recommended over A1C at 4–12 weeks postpartum, because A1C may be persistently impacted (lowered) by the increased red blood cell turnover related to pregnancy, by blood loss at delivery, or by the preceding 3-month glucose profile. The OGTT is more sensitive at detecting glucose intolerance, including both prediabetes and diabetes, and has been examined as a screening tool for these conditions in the first 12 weeks after delivery in individuals who had a recent pregnancy with GDM (188,189). OGTT testing immediately postpartum, while still hospitalized, has demonstrated improved engagement in testing but also variably reduced sensitivity to the diagnosis of impaired fasting glucose, impaired glucose tolerance, and type 2 diabetes (190,191). In a small subset of individuals who are unable to tolerate or who decline the postpartum glucose

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tolerance test, an A1C performed at 6–12 months postpartum may be considered. A retrospective study aimed to evaluate the performance of A1C as a postpartum screening test (192). The study included 677 out of 9,118 individuals with a recent GDM diagnosis who had both A1C and 2-h 75-g OGTT testing performed for clinical purposes between delivery and 12 months postpartum. A1C level of 5.7% had 80% sensitivity and 80.8% specificity for the detection of a 2-h plasma glucose value ≥200 mg/dL (≥11.1 mmol/L). A weakness of this study is the likelihood of selection bias, as it is not standard of care to perform both tests unless there is higher concern for a diagnosis of diabetes.

Individuals with a history of GDM should have ongoing screening for prediabetes or type 2 diabetes every 1–3 years, even if results of the initial 75-g OGTT at 4–12 weeks postpartum are normal. Ongoing evaluation may be performed with any recommended glycemic test (see section 2, "Diagnosis and Classification of Diabetes").

Individuals with a history of GDM have an increased lifetime maternal risk for diabetes estimated at 50–60% (193,194), and those with GDM have a 10-fold increased risk of developing type 2 diabetes compared with those without GDM (193). Absolute risk of developing type 2 diabetes after GDM increases linearly through a person's lifetime, being ∼20% at 10 years, 30% at 20 years, 40% at 30 years, 50% at 40 years, and 60% at 50 years (194). Hazard ratios for incident diabetes were significantly elevated for a history of GDM in a single pregnancy but were even higher for a history of two GDM pregnancies in a large retrospective cohort study (hazard ratios ranged from 4.35- to 15.8-fold based on number of pregnancies with GDM and which pregnancy the individual had GDM (first or second) (195). In the prospective Nurses' Health Study II (NHS II), subsequent diabetes risk after a history of GDM was significantly lower in those who followed healthy eating patterns (196). Adjusting for BMI moderately attenuated this association. Interpregnancy weight gain is associated with increased risk of adverse pregnancy outcomes (197) and higher risk of GDM, while in people with BMI >25 kg/m^2, weight loss is associated with lower risk of developing GDM in the subsequent pregnancy (198). Development of type 2 diabetes is 18% higher per unit of BMI increase from prepregnancy BMI at follow-up, highlighting the importance of effective weight management after GDM (199). In addition, postdelivery lifestyle interventions are effective in reducing risk of type 2 diabetes (200).

Both metformin and intensive lifestyle intervention prevent or delay progression to diabetes in individuals with prediabetes and a history of GDM. Only five to six individuals with prediabetes and a history of GDM need to be treated with either intervention to prevent one case of diabetes over 3 years (201). In these individuals, lifestyle intervention and metformin reduced progression to diabetes by 35% and 40%, respectively, over 10 years compared with placebo (202). If pregnancy has motivated the adoption of healthy nutrition, building on these gains to support weight loss is recommended in the postpartum period (see section 3, "Prevention or Delay of Diabetes and Associated Comorbidities").

References

1. Dabelea D, Hanson RL, Lindsay RS, et al. Intrauterine exposure to diabetes conveys risks for type 2 diabetes and obesity: a study of discordant sibships. Diabetes 2000;49:2208–2211
2. Holmes VA, Young IS, Patterson CC, et al.; Diabetes and Pre-eclampsia Intervention Trial Study Group. Optimal glycemic control, pre-eclampsia, and gestational hypertension in women with type 1 diabetes in the diabetes and pre-eclampsia intervention trial. Diabetes Care 2011; 34:1683–1688
3. Wahabi HA, Fayed A, Esmaeil S, et al. Systematic review and meta-analysis of the effectiveness of pre-pregnancy care for women with diabetes for improving maternal and perinatal outcomes. PLoS One 2020;15:e0237571
4. Charron-Prochownik D, Sereika SM, Becker D, et al. Long-term effects of the booster-enhanced READY-Girls preconception counseling program on intentions and behaviors for family planning in teens with diabetes. Diabetes Care 2013;36: 3870–3874
5. ACOG Committee Opinion No. 762. Prepregnancy Counseling. Obstet Gynecol 2019; 133:e78–e89
6. Peterson C, Grosse SD, Li R, et al. Preventable health and cost burden of adverse birth outcomes associated with pregestational diabetes in the United States. Am J Obstet Gynecol 2015;212: 74.e71–74.e9
7. Britton LE, Hussey JM, Berry DC, Crandell JL, Brooks JL, Bryant AG. Contraceptive use among women with prediabetes and diabetes in a US national sample. J Midwifery Womens Health 2019;64:36–45
8. Morris JR, Tepper NK. Description and comparison of postpartum use of effective contraception among women with and without diabetes. Contraception 2019;100:474–479
9. Goldstuck ND, Steyn PS. The intrauterine device in women with diabetes mellitus type I and II: a systematic review. ISRN Obstet Gynecol 2013;2013:814062
10. Wu JP, Moniz MH, Ursu AN. Long-acting reversible contraception-highly efficacious, safe, and underutilized. JAMA 2018;320:397–398
11. American College of Obstetricians and Gynecologists' Committee on Practice Bulletins—Obstetrics. ACOG Practice Bulletin No. 201: Pregestational Diabetes Mellitus. Obstet Gynecol 2018;132:e228–e248
12. Centers for Disease Control and Prevention. U.S. Medical Eligibility Criteria for Contraceptive Use, 2024, 2024. Accessed 15 August 2025. Available from https://www.cdc.gov/contraception/hcp/usmec/index.html
13. Guerin A, Nisenbaum R, Ray JG. Use of maternal GHb concentration to estimate the risk of congenital anomalies in the offspring of women with prepregnancy diabetes. Diabetes Care 2007;30:1920–1925
14. Jensen DM, Korsholm L, Ovesen P, et al. Periconceptional A1C and risk of serious adverse pregnancy outcome in 933 women with type 1 diabetes. Diabetes Care 2009;32:1046–1048
15. Nielsen GL, Møller M, Sørensen HT. HbA1c in early diabetic pregnancy and pregnancy outcomes: a Danish population-based cohort study of 573 pregnancies in women with type 1 diabetes. Diabetes Care 2006;29:2612–2616
16. Ludvigsson JF, Neovius M, Söderling J, et al. Maternal glycemic control in type 1 diabetes and the risk for preterm birth: a population-based cohort study. Ann Intern Med 2019;170:691–701
17. Ludvigsson JF, Neovius M, Söderling J, et al. Periconception glycaemic control in women with type 1 diabetes and risk of major birth defects: population based cohort study in Sweden. Bmj 2018;362:k2638
18. Wahabi HA, Alzeidan RA, Bawazeer GA, Alansari LA, Esmaeil SA. Preconception care for diabetic women for improving maternal and fetal outcomes: a systematic review and meta-analysis. BMC Pregnancy Childbirth 2010;10:63
19. American Diabetes Association. Diabetes and Reproductive Health for Girls, 2016. Accessed 15 August 2025. Available from https://diabetes.org/sites/dpro/files/2025-02/16-ready-girls-book-proof-4-15-16.pdf
20. ACOG Practice Bulletin No. 190. Gestational Diabetes Mellitus. Obstet Gynecol 2018;131: e49–e64
21. Metzger BE, Lowe LP, Dyer AR, et al.; HAPO Study Cooperative Research Group. Hyperglycemia and adverse pregnancy outcomes. N Engl J Med 2008;358:1991–2002
22. Scholtens DM, Kuang A, Lowe LP, et al.; HAPO Follow-Up Study Cooperative Research Group. Hyperglycemia and Adverse Pregnancy Outcome Follow-up Study (HAPO FUS): maternal glycemia and childhood glucose metabolism. Diabetes Care 2019;42:381–392
23. Lowe WL, Scholtens DM, Kuang A, et al.; HAPO Follow-up Study Cooperative Research Group. Hyperglycemia and Adverse Pregnancy Outcome Follow-up Study (HAPO FUS): maternal gestational diabetes mellitus and childhood glucose metabolism. Diabetes Care 2019;42:372–380
24. Obesity in pregnancy: ACOG Practice Bulletin, Number 230. Obstet Gynecol 2021;137:e128–e144
25. Phelan S, Jelalian E, Coustan D, et al. Randomized controlled trial of prepregnancy lifestyle intervention to reduce recurrence of

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gestational diabetes mellitus. Am J Obstet Gynecol 2023;229:158.e151-158–158.e14
26. Clausen TD, Mathiesen E, Ekbom P, Hellmuth E, Mandrup-Poulsen T, Damm P. Poor pregnancy outcome in women with type 2 diabetes. Diabetes Care 2005;28:323–328
27. Cundy T, Gamble G, Neale L, et al. Differing causes of pregnancy loss in type 1 and type 2 diabetes. Diabetes Care 2007;30:2603–2607
28. Barry MJ, Nicholson WK, Silverstein M, et al.; US Preventive Services Task Force. Folic acid supplementation to prevent neural tube defects: US Preventive Services Task Force reaffirmation recommendation statement. JAMA 2023;330: 454–459
29. Alexander EK, Pearce EN, Brent GA, et al. 2017 Guidelines of the American Thyroid Association for the diagnosis and management of thyroid disease during pregnancy and the postpartum. Thyroid 2017; 27:315–389
30. Ramos DE. Preconception health: changing the paradigm on well-woman health. Obstet Gynecol Clin North Am 2019;46:399–408
31. American Diabetes Association. Preconception care of women with diabetes. Diabetes Care 2004; 27(Suppl 1):S76–S78
32. Relph S, Patel T, Delaney L, Sobhy S, Thangaratinam S. Adverse pregnancy outcomes in women with diabetes-related microvascular disease and risks of disease progression in pregnancy: a systematic review and meta-analysis. PLoS Med 2021;18:e1003856
33. Bullo M, Tschumi S, Bucher BS, Bianchetti MG, Simonetti GD. Pregnancy outcome following exposure to angiotensin-converting enzyme inhibitors or angiotensin receptor antagonists: a systematic review. Hypertension 2012;60:444–450
34. Agarwala A, Dixon DL, Gianos E, et al. Dyslipidemia management in women of reproductive potential: an expert clinical consensus from the National Lipid Association. J Clin Lipidol 2024; 18:e664–e684
35. Novo Nordisk. Semaglutide prescribing information. 2023. Accessed 15 August 2025. Available from https://www.novo-pi.com/ozempic.pdf
36. Eli Lilly Canada Inc. Tirzepatide product monograph. 2022. Accessed 15 August 2025. Available from https://pdf.hres.ca/dpd_pm/00068421.PDF
37. Widyaputri F, Rogers S, Lim L. Global estimates of diabetic retinopathy prevalence and progression in pregnant individuals with preexisting diabetes: a meta-analysis. JAMA Ophthalmol 2022; 140:1137–1138
38. García-Patterson A, Gich I, Amini SB, Catalano PM, de Leiva A, Corcoy R. Insulin requirements throughout pregnancy in women with type 1 diabetes mellitus: three changes of direction. Diabetologia 2010;53:446–451
39. Mathiesen JM, Secher AL, Ringholm L, et al. Changes in basal rates and bolus calculator settings in insulin pumps during pregnancy in women with type 1 diabetes. J Matern Fetal Neonatal Med 2014;27:724–728
40. Diabetes Technology Network UK. Best Practice Guide: Using Diabetes Technology in Pregnancy, 2020. Accessed 15 August 2025. Available from https://abcd.care/sites/abcd.care/files/site_uploads/Resources/DTN/BP-Pregnancy-DTN-V2.0.pdf
41. de Veciana M, Major CA, Morgan MA, et al. Postprandial versus preprandial blood glucose monitoring in women with gestational diabetes mellitus requiring insulin therapy. N Engl J Med 1995;333:1237–1241
42. Jovanovic-Peterson L, Peterson CM, Reed GF, et al. Maternal postprandial glucose levels and infant birth weight: the Diabetes in Early Pregnancy Study. The National Institute of Child Health and Human Development-Diabetes in Early Pregnancy Study. Am J Obstet Gynecol 1991; 164:103–111
43. Hernandez TL, Friedman JE, Van Pelt RE, Barbour LA. Patterns of glycemia in normal pregnancy: should the current therapeutic targets be challenged? Diabetes Care 2011;34:1660–1668
44. Battelino T, Danne T, Bergenstal RM, et al. Clinical targets for continuous glucose monitoring data interpretation: recommendations from the International Consensus on Time in Range. Diabetes Care 2019;42:1593–1603
45. Metzger BE, Buchanan TA, Coustan DR, et al. Summary and recommendations of the Fifth International Workshop-Conference on Gestational Diabetes Mellitus. Diabetes Care 2007;30(Suppl 2):S251–S260
46. ACOG Clinical Practice Update: Screening for Gestational and Pregestational Diabetes in Pregnancy and Postpartum. Obstet Gynecol 2024;144:e20–e23
47. Sweeting A, Hare MJ, de Jersey SJ, et al. Australasian Diabetes in Pregnancy Society (ADIPS) 2025 consensus recommendations for the screening, diagnosis and classification of gestational diabetes. Med J Aust 2025;223:161–167
48. Zhu W-W, Yang H-X, Wei Y-M, et al. Evaluation of the value of fasting plasma glucose in the first prenatal visit to diagnose gestational diabetes mellitus in China. Diabetes Care 2013; 36:586–590
49. Kattini R, Hummelen R, Kelly L. Early gestational diabetes mellitus screening with glycated hemoglobin: a systematic review. J Obstet Gynaecol Can 2020;42:1379–1384
50. Mañé L, Flores-Le Roux JA, Benaiges D, et al. Role of first-trimester HbA1c as a predictor of adverse obstetric outcomes in a multiethnic cohort. J Clin Endocrinol Metab 2017;102:390–397
51. Hughes RCE, Moore MP, Gullam JE, Mohamed K, Rowan J. An early pregnancy HbA1c ≥5.9% (41 mmol/mol) is optimal for detecting diabetes and identifies women at increased risk of adverse pregnancy outcomes. Diabetes Care 2014;37:2953–2959
52. Shen L, Zhang S, Wen J, et al. Universal screening for hyperglycemia in early pregnancy and the risk of adverse pregnancy outcomes. BMC Pregnancy Childbirth 2025;25:203
53. Simmons D, Immanuel J, Hague WM, et al.; TOBOGM Research Group. Treatment of gestational diabetes mellitus diagnosed early in pregnancy. N Engl J Med 2023;388:2132–2144
54. Simmons D, Immanuel J, Hague WM, et al.; TOBOGM Research Group. Regression from early GDM to normal glucose tolerance and adverse pregnancy outcomes in the Treatment of Booking Gestational Diabetes Mellitus study. Diabetes Care 2024;47:2079–2084
55. Sweeting A, Enticott J, Immanuel J, et al.; TOBOGM Research Group. Relationship between early-pregnancy glycemia and adverse outcomes: findings from the TOBOGM study. Diabetes Care 2024;47:2085–2092
56. Simmons D, Immanuel J, Hague WM, et al.; TOBOGM Research Group. Perinatal outcomes in early and late gestational diabetes mellitus after treatment from 24-28 weeks' gestation: a TOBOGM secondary analysis. Diabetes Care 2024;47: 2093–2101
57. García-Patterson A, Balsells M, Solà I, Corcoy R. Detection and treatment of early gestational diabetes mellitus: a systematic review and meta-analysis of randomized controlled trials. Am J Obstet Gynecol 2025:S0002
58. Bhattacharya S, Nagendra L, Dutta D, Kamrul-Hasan ABM. Treatment versus observation in early gestational diabetes mellitus: a systematic review and meta-analysis of randomized controlled trials. J Clin Endocrinol Metab 2025;110:1781–1791
59. Ho Y-R, Wang P, Lu M-C, Tseng S-T, Yang C-P, Yan Y-H. Associations of mid-pregnancy HbA1c with gestational diabetes and risk of adverse pregnancy outcomes in high-risk Taiwanese women. PLoS One 2017;12:e0177563
60. Maresh MJA, Holmes VA, Patterson CC, et al.; Diabetes and Pre-eclampsia Intervention Trial Study Group. Glycemic targets in the second and third trimester of pregnancy for women with type 1 diabetes. Diabetes Care 2015;38:34–42
61. Nielsen LR, Ekbom P, Damm P, et al. HbA_{1c} levels are significantly lower in early and late pregnancy. Diabetes Care 2004;27:1200–1201
62. Mosca A, Paleari R, Dalfrà MG, et al. Reference intervals for hemoglobin A1c in pregnant women: data from an Italian multicenter study. Clin Chem 2006;52:1138–1143
63. Hummel M, Marienfeld S, Huppmann M, et al. Fetal growth is increased by maternal type 1 diabetes and HLA DR4-related gene interactions. Diabetologia 2007;50:850–858
64. Cyganek K, Skupien J, Katra B, et al. Risk of macrosomia remains glucose-dependent in a cohort of women with pregestational type 1 diabetes and good glycemic control. Endocrine 2017;55:447–455
65. Abell SK, Boyle JA, de Courten B, et al. Impact of type 2 diabetes, obesity and glycaemic control on pregnancy outcomes. Aust N Z J Obstet Gynaecol 2017;57:308–314
66. Temple RC, Aldridge V, Stanley K, Murphy HR. Glycaemic control throughout pregnancy and risk of pre-eclampsia in women with type I diabetes. BJOG 2006;113:1329–1332
67. Combs CA, Gunderson E, Kitzmiller JL, Gavin LA, Main EK. Relationship of fetal macrosomia to maternal postprandial glucose control during pregnancy. Diabetes Care 1992;15:1251–1257
68. Langer O, Levy J, Brustman L, Anyaegbunam A, Merkatz R, Divon M. Glycemic control in gestational diabetes mellitus–how tight is tight enough: small for gestational age versus large for gestational age? Am J Obstet Gynecol 1989;161: 646–653
69. Feig DS, Donovan LE, Corcoy R, et al.; CONCEPTT Collaborative Group. Continuous Glucose Monitoring in Pregnant Women with Type 1 Diabetes (CONCEPTT): a multicentre international randomised controlled trial. Lancet 2017;390: 2347–2359
70. Kristensen K, Ögge LE, Sengpiel V, et al. Continuous glucose monitoring in pregnant women with type 1 diabetes: an observational cohort study of 186 pregnancies. Diabetologia 2019;62:1143–1153
71. Law GR, Gilthorpe MS, Secher AL, et al. Translating HbA1c measurements into estimated

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average glucose values in pregnant women with diabetes. Diabetologia 2017;60:618–624
72. Sanusi AA, Xue Y, McIlwraith C, et al. Association of continuous glucose monitoring metrics with pregnancy outcomes in patients with preexisting diabetes. Diabetes Care 2024;47:89–96
73. Sobhani NC, Goemans S, Nguyen A, et al. Continuous glucose monitoring in pregnancies with type 1 diabetes: small increases in time-in-range improve maternal and perinatal outcomes. Am J Obstet Gynecol 2024;231:467.e1–467.e8
74. Ahmed RJ, Gafni A, Hutton EK, et al.; CONCEPTT Collaborative Group. The cost implications of continuous glucose monitoring in pregnant women with type 1 diabetes in 3 Canadian provinces: a posthoc cost analysis of the CONCEPTT trial. CMAJ Open 2021;9:E627–E634
75. Szmuilowicz ED, Barbour L, Brown FM, et al. Continuous glucose monitoring metrics for pregnancies complicated by diabetes: critical appraisal of current evidence. J Diabetes Sci Technol 2024;18:819–834
76. Wei Q, Sun Z, Yang Y, Yu H, Ding H, Wang S. Effect of a CGMS and SMBG on maternal and neonatal outcomes in gestational diabetes mellitus: a randomized controlled trial. Sci Rep 2016;6:19920
77. Burk J, Ross GP, Hernandez TL, Colagiuri S, Sweeting A. Evidence for improved glucose metrics and perinatal outcomes with continuous glucose monitoring compared to self-monitoring in diabetes during pregnancy. Am J Obstet Gynecol 2025;233:162–175
78. Lai M, Weng J, Yang J, et al. Effect of continuous glucose monitoring compared with self-monitoring of blood glucose in gestational diabetes patients with HbA1c<6%: a randomized controlled trial. Front Endocrinol 2023;14:1174239
79. Amylidi-Mohr S, Zennaro G, Schneider S, Raio L, Mosimann B, Surbek D. Continuous glucose monitoring in the management of gestational diabetes in Switzerland (DipGluMo): an open-label, single-centre, randomised, controlled trial. Lancet Diabetes Endocrinol 2025;13:591–599
80. Valent AM, Rickert M, Pagan CH, Ward L, Dunn E, Rincon M. Real-time continuous glucose monitoring in pregnancies with gestational diabetes mellitus: a randomized controlled trial. Diabetes Care 2025;48:1581–1588
81. Polsky S, Valent AM, Isganaitis E, et al. Performance of the Dexcom G7 continuous glucose monitoring system in pregnant women with diabetes. Diabetes Technol Ther 2024;26:307–312
82. Neoh SL, Grisoni JA, Feig DS, Murphy HR; CONCEPTT Collaborative Group. Dietary intakes of women with type 1 diabetes before and during pregnancy: a pre-specified secondary subgroup analysis among CONCEPTT participants. Diabet Med 2020;37:1841–1848
83. Marshall NE, Abrams B, Barbour LA, et al. The importance of nutrition in pregnancy and lactation: lifelong consequences. Am J Obstet Gynecol 2022;226:607–632
84. U.S. Department of Agriculture and U.S. Department of Health and Human Services. *Dietary Guidelines for Americans, 2020–2025*. Accessed 15 August 2025. Available from https://www.dietaryguidelines.gov/sites/default/files/2020-12/Dietary_Guidelines_for_Americans_2020-2025.pdf
85. Medeiros DM. In *Dietary Reference Intakes: The Essential Guide to Nutrient Requirements*. Otten JJ, Hellwig JP, Meyers LD, Eds. Washington, DC, National Academies Press, 2006
86. Hernandez TL, Mande A, Barbour LA. Nutrition therapy within and beyond gestational diabetes. Diabetes Res Clin Pract 2018;145:39–50
87. Hernandez TL, Van Pelt RE, Anderson MA, et al. A higher-complex carbohydrate diet in gestational diabetes mellitus achieves glucose targets and lowers postprandial lipids: a randomized crossover study. Diabetes Care 2014;37:1254–1262
88. Han S, Middleton P, Shepherd E, Van Ryswyk E, Crowther CA. Different types of dietary advice for women with gestational diabetes mellitus. Cochrane Database Syst Rev 2017;2:CD009275
89. Viana LV, Gross JL, Azevedo MJ. Dietary intervention in patients with gestational diabetes mellitus: a systematic review and meta-analysis of randomized clinical trials on maternal and newborn outcomes. Diabetes Care 2014;37:3345–3355
90. *Weight Gain During Pregnancy: Reexamining the Guidelines*. Rasmussen KM, Yaktine AL, Eds. Washington, DC, 2009. Available from https://nap.nationalacademies.org/catalog/12584/weight-gain-during-pregnancy-reexamining-the-guidelines
91. Koivusalo SB, Rönö K, Klemetti MM, et al. Erratum. Gestational diabetes mellitus can be prevented by lifestyle intervention: the Finnish Gestational Diabetes Prevention Study (RADIEL). A randomized controlled trial. Diabetes Care 2016;39:24–30. Diabetes Care 2017;40:1133
92. Wang C, Wei Y, Zhang X, et al. A randomized clinical trial of exercise during pregnancy to prevent gestational diabetes mellitus and improve pregnancy outcome in overweight and obese pregnant women. Am J Obstet Gynecol 2017;216:340–351
93. Griffith RJ, Alsweiler J, Moore AE, et al. Interventions to prevent women from developing gestational diabetes mellitus: an overview of Cochrane Reviews. Cochrane Database Syst Rev 2020;6:Cd012394
94. Mayo K, Melamed N, Vandenberghe H, Berger H. The impact of adoption of the international association of diabetes in pregnancy study group criteria for the screening and diagnosis of gestational diabetes. Am J Obstet Gynecol 2015;212:224.e1–e221
95. U.S. Department of Health and Human Services. Physical Activity Guidelines for Americans, 2nd edition. 2018. Accessed 15 August 2025. Available from https://health.gov/sites/default/files/2019-09/Physical_Activity_Guidelines_2nd_edition.pdf
96. Davenport MH, Ruchat S-M, Poitras VJ, et al. Prenatal exercise for the prevention of gestational diabetes mellitus and hypertensive disorders of pregnancy: a systematic review and meta-analysis. Br J Sports Med 2018;52:1367–1375
97. Davenport MH, Sobierajski F, Mottola MF, et al. Glucose responses to acute and chronic exercise during pregnancy: a systematic review and meta-analysis. Br J Sports Med 2018;52:1357–1366
98. Laredo-Aguilera JA, Gallardo-Bravo M, Rabanales-Sotos JA, Cobo-Cuenca AI, Carmona-Torres JM. Physical activity programs during pregnancy are effective for the control of gestational diabetes mellitus. Int J Environ Res Public Health 2020;17:6151
99. Xie W, Dai P, Qin Y, Wu M, Yang B, Yu X. Effectiveness of telemedicine for pregnant women with gestational diabetes mellitus: an updated meta-analysis of 32 randomized controlled trials with trial sequential analysis. BMC Pregnancy Childbirth 2020;20:198
100. Padmanabhan S, Lee VW, Mclean M, et al. The association of falling insulin requirements with maternal biomarkers and placental dysfunction: a prospective study of women with preexisting diabetes in pregnancy. Diabetes Care 2017;40:1323–1330
101. Ram M, Feinmesser L, Shinar S, Maslovitz S. The importance of declining insulin requirements during pregnancy in patients with pre-gestational gestational diabetes mellitus. Eur J Obstet Gynecol Reprod Biol 2017;215:148–152
102. Pollex EK, Feig DS, Lubetsky A, Yip PM, Koren G. Insulin glargine safety in pregnancy: a transplacental transfer study. Diabetes Care 2010;33:29–33
103. Holcberg G, Tsadkin-Tamir M, Sapir O, et al. Transfer of insulin lispro across the human placenta. Eur J Obstet Gynecol Reprod Biol 2004;115:117–118
104. Boskovic R, Feig DS, Derewlany L, Knie B, Portnoi G, Koren G. Transfer of insulin lispro across the human placenta: in vitro perfusion studies. Diabetes Care 2003;26:1390–1394
105. McCance DR, Damm P, Mathiesen ER, et al. Evaluation of insulin antibodies and placental transfer of insulin aspart in pregnant women with type 1 diabetes mellitus. Diabetologia 2008;51:2141–2143
106. Farrar D, Tuffnell DJ, West J, West HM. Continuous subcutaneous insulin infusion versus multiple daily injections of insulin for pregnant women with diabetes. Cochrane Database Syst Rev 2016;2016:CD005542
107. Suffecool K, Rosenn B, Niederkofler EE, et al. Insulin detemir does not cross the human placenta. Diabetes Care 2015;38:e20–e21
108. Mathiesen ER, Alibegovic AC, Corcoy R, et al.; EXPECT Study Group. Insulin degludec versus insulin detemir, both in combination with insulin aspart, in the treatment of pregnant women with type 1 diabetes (EXPECT): an open-label, multinational, randomised, controlled, non-inferiority trial. Lancet Diabetes Endocrinol 2023;11:86–95
109. O'Neill SM, Kenny LC, Khashan AS, West HM, Smyth RM, Kearney PM. Different insulin types and regimens for pregnant women with pre-existing diabetes. Cochrane Database Syst Rev 2017;2:Cd011880
110. Fishel Bartal M, Ward C, Blackwell SC, et al. Detemir vs neutral protamine Hagedorn insulin for diabetes mellitus in pregnancy: a comparative effectiveness, randomized controlled trial. Am J Obstet Gynecol 2021;225:87 e81–87 e10
111. Pollex E, Moretti ME, Koren G, Feig DS. Safety of insulin glargine use in pregnancy: a systematic review and meta-analysis. Ann Pharmacother 2011;45:9–16
112. Novo Nordisk. Insulin degludec prescribing information, 2025. Accessed 15 August 2025. Available from https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/203314s018s020lbl.pdf
113. Hirsch IB. Insulin analogues. N Engl J Med 2005;352:174–183
114. Hod M, Damm P, Kaaja R, et al.; Insulin Aspart Pregnancy Study Group. Fetal and perinatal outcomes in type 1 diabetes pregnancy: a randomized study comparing insulin aspart

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with human insulin in 322 subjects. Am J Obstet Gynecol 2008;198:186.e181–186.e187
115. Mathiesen ER, Kinsley B, Amiel SA, et al.; Insulin Aspart Pregnancy Study Group. Maternal glycemic control and hypoglycemia in type 1 diabetic pregnancy: a randomized trial of insulin aspart versus human insulin in 322 pregnant women. Diabetes Care 2007;30:771–776
116. Nørgaard SK, Søholm JC, Mathiesen ER, et al. Faster-acting insulin aspart versus insulin aspart in the treatment of type 1 or type 2 diabetes during pregnancy and post-delivery (CopenFast): an open-label, single-centre, randomised controlled trial. Lancet Diabetes Endocrinol 2023;11: 811–821
117. Søholm JC, Nørgaard SK, Nørgaard K, et al. Sensor-derived glycaemic metrics in pregnant women with type 1 diabetes randomised to faster acting insulin aspart or insulin aspart—a secondary analysis of the CopenFast trial. Diabet Med 2025;42:e15467
118. Kernaghan D, Farrell T, Hammond P, Owen P. Fetal growth in women managed with insulin pump therapy compared to conventional insulin. Eur J Obstet Gynecol Reprod Biol 2008;137:47–49
119. Lee TTM, Collett C, Bergford S, et al.; AiDAPT Collaborative Group. Automated insulin delivery in women with pregnancy complicated by type 1 diabetes. N Engl J Med 2023;389:1566–1578
120. Lawton J, Kimbell B, Closs M, et al. Listening to women: experiences of using closed-loop in type 1 diabetes pregnancy. Diabetes Technol Ther 2023;25:845–855
121. Benhalima K, Beunen K, Van Wilder N, et al. Comparing advanced hybrid closed loop therapy and standard insulin therapy in pregnant women with type 1 diabetes (CRISTAL): a parallel-group, open-label, randomised controlled trial. Lancet Diabetes Endocrinol 2024;12:390–403
122. Polsky S, Buschur E, Dungan K, et al. Randomized trial of assisted hybrid closed-loop therapy versus sensor-augmented pump therapy in pregnancy. Diabetes Technol Ther 2024;26: 547–555
123. Szmuilowicz ED, Levy CJ, Buschur EO, Polsky S. Expert guidance on off-label use of hybrid closed-loop therapy in pregnancies complicated by diabetes. Diabetes Technol Ther 2023;25:363–373
124. Wilkie GL, Delpapa E, Leftwich HK. Intrapartum continuous subcutaneous insulin infusion vs intravenous insulin infusion among pregnant individuals with type 1 diabetes mellitus: a randomized controlled trial. Am J Obstet Gynecol 2023;229:680.e681–680.e688
125. Donovan LE, Feig DS, Lemieux P, et al. A randomized trial of closed-loop insulin delivery postpartum in type 1 diabetes. Diabetes Care 2023;46:2258–2266
126. Beunen K, Gillard P, Van Wilder N, et al. Advanced hybrid closed-loop therapy compared with standard insulin therapy intrapartum and early postpartum in women with type 1 diabetes: a secondary observational analysis from the CRISTAL randomized controlled trial. Diabetes Care 2024;47:2002–2011
127. Lee TTM, Collett C, Bergford S, et al.; AiDAPT Collaborative Group. Automated insulin delivery during the first 6 months postpartum (AiDAPT): a prespecified extension study. Lancet Diabetes Endocrinol 2025;13:210–220
128. Szmuilowicz ED, Durnwald C, Feig DS. Practical approach to continuous glucose monitoring (CGM) interpretation and automated insulin delivery (AID) use in pregnancy: considerations for obstetric providers. J Diabetes Sci Technol 2025:19322968 251330651
129. Hartling L, Dryden DM, Guthrie A, Muise M, Vandermeer B, Donovan L. Benefits and harms of treating gestational diabetes mellitus: a systematic review and meta-analysis for the U.S. Preventive Services Task Force and the National Institutes of Health Office of Medical Applications of Research. Ann Intern Med 2013;159:123–129
130. Rowan JA, Hague WM, Gao W, Battin MR, Moore MP; MiG Trial Investigators. Metformin versus insulin for the treatment of gestational diabetes. N Engl J Med 2008;358:2003–2015
131. Gui J, Liu Q, Feng L. Metformin vs insulin in the management of gestational diabetes: a meta-analysis. PLoS One 2013;8:e64585
132. Song R, Chen L, Chen Y, et al. Comparison of glyburide and insulin in the management of gestational diabetes: a meta-analysis. PLoS One 2017;12:e0182488
133. Hebert MF, Ma X, Naraharisetti SB, et al.; Obstetric-Fetal Pharmacology Research Unit Network. Are we optimizing gestational diabetes treatment with glyburide? The pharmacologic basis for better clinical practice. Clin Pharmacol Ther 2009;85:607–614
134. Malek R, Davis SN. Pharmacokinetics, efficacy and safety of glyburide for treatment of gestational diabetes mellitus. Expert Opin Drug Metab Toxicol 2016;12:691–699
135. Balsells M, García-Patterson A, Solà I, Roqué M, Gich I, Corcoy R. Glibenclamide, metformin, and insulin for the treatment of gestational diabetes: a systematic review and meta-analysis. Bmj 2015;350:h102
136. Tarry-Adkins JL, Aiken CE, Ozanne SE. Comparative impact of pharmacological treatments for gestational diabetes on neonatal anthropometry independent of maternal glycaemic control: a systematic review and meta-analysis. PLoS Med 2020;17:e1003126
137. Sénat M-V, Affres H, Letourneau A, et al.; Groupe de Recherche en Obstétrique et Gynécologie (GROG). Effect of glyburide vs subcutaneous insulin on perinatal complications among women with gestational diabetes: a randomized clinical trial. JAMA 2018;319:1773–1780
138. Rademaker D, de Wit L, Duijnhoven RG, et al.; SUGAR-DIP Study Group. Oral glucose-lowering agents vs insulin for gestational diabetes: a randomized clinical trial. JAMA 2025;333: 470–478
139. Silva JC, Pacheco C, Bizato J, de Souza BV, Ribeiro TE, Bertini AM. Metformin compared with glyburide for the management of gestational diabetes. Int J Gynaecol Obstet 2010;111:37–40
140. Nachum Z, Zafran N, Salim R, et al. Glyburide versus metformin and their combination for the treatment of gestational diabetes mellitus: a randomized controlled study. Diabetes Care 2017;40:332–337
141. Jiang Y-F, Chen X-Y, Ding T, Wang X-F, Zhu Z-N, Su S-W. Comparative efficacy and safety of OADs in management of GDM: network meta-analysis of randomized controlled trials. J Clin Endocrinol Metab 2015;100:2071–2080
142. Dunne F, Newman C, Alvarez-Iglesias A, et al. Early metformin in gestational diabetes: a randomized clinical trial. JAMA 2023;330:1547–1556
143. Doi SAR, Furuya-Kanamori L, Toft E, et al. Metformin in pregnancy to avert gestational diabetes in women at high risk: meta-analysis of randomized controlled trials. Obes Rev 2020;21: e12964
144. Feig DS, Donovan LE, Zinman B, et al.; MiTy Collaborative Group. Metformin in women with type 2 diabetes in pregnancy (MiTy): a multicentre, international, randomised, placebo-controlled trial. Lancet Diabetes Endocrinol 2020;8:834–844
145. Boggess KA, Valint A, Refuerzo JS, et al. Metformin plus insulin for preexisting diabetes or gestational diabetes in early pregnancy: the MOMPOD randomized clinical trial. JAMA 2023; 330:2182–2190
146. Vanky E, Stridsklev S, Heimstad R, et al. Metformin versus placebo from first trimester to delivery in polycystic ovary syndrome: a randomized, controlled multicenter study. J Clin Endocrinol Metab 2010;95:E448–455
147. Palomba S, Orio F, Falbo A, et al. Prospective parallel randomized, double-blind, double-dummy controlled clinical trial comparing clomiphene citrate and metformin as the first-line treatment for ovulation induction in nonobese anovulatory women with polycystic ovary syndrome. J Clin Endocrinol Metab 2005;90:4068–4074
148. Palomba S, Orio F, Nardo LG, et al. Metformin administration versus laparoscopic ovarian diathermy in clomiphene citrate-resistant women with polycystic ovary syndrome: a prospective parallel randomized double-blind placebo-controlled trial. J Clin Endocrinol Metab 2004;89:4801–4809
149. Legro RS, Barnhart HX, Schlaff WD, et al.; Cooperative Multicenter Reproductive Medicine Network. Clomiphene, metformin, or both for infertility in the polycystic ovary syndrome. N Engl J Med 2007;356:551–566
150. Vanky E, Zahlsen K, Spigset O, Carlsen SM. Placental passage of metformin in women with polycystic ovary syndrome. Fertil Steril 2005;83: 1575–1578
151. Charles B, Norris R, Xiao X, Hague W. Population pharmacokinetics of metformin in late pregnancy. Ther Drug Monit 2006;28:67–72
152. Bolte E, Dean T, Garcia B, et al. Initiation of metformin in early pregnancy results in fetal bioaccumulation, growth restriction, and renal dysmorphology in a primate model. Am J Obstet Gynecol 2024;231:352.e1–352.e16
153. Rowan JA, Rush EC, Plank LD, et al. Metformin in gestational diabetes: the offspring follow-up (MiG TOFU): body composition and metabolic outcomes at 7-9 years of age. BMJ Open Diabetes Res Care 2018;6:e000456
154. Hanem LGE, Stridsklev S, Júlíusson PB, et al. Metformin use in PCOS pregnancies increases the risk of offspring overweight at 4 years of age: follow-up of two RCTs. J Clin Endocrinol Metab 2018;103:1612–1621
155. Tarry-Adkins JL, Aiken CE, Ozanne SE. Neonatal, infant, and childhood growth following metformin versus insulin treatment for gestational diabetes: a systematic review and meta-analysis. PLoS Med 2019;16:e1002848
156. Hanem LGE, Salvesen Ø, Juliusson PB, et al. Intrauterine metformin exposure and offspring cardiometabolic risk factors (PedMet study): a 5-10 year follow-up of the PregMet randomised controlled trial. Lancet Child Adolesc Health 2019;3:166–174

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157. Feig DS, Sanchez JJ, Murphy KE, et al.; MiTy Kids Collaborative Group. Outcomes in children of women with type 2 diabetes exposed to metformin versus placebo during pregnancy (MiTy Kids): a 24-month follow-up of the MiTy randomised controlled trial. Lancet Diabetes Endocrinol 2023;11:191–202
158. Wyckoff JA, Lapolla A, Asias-Dinh BD, et al. Preexisting diabetes and pregnancy: an Endocrine Society and European Society of Endocrinology joint clinical practice guideline. J Clin Endocrinol Metab 2025;110:2405–2452
159. Chiu Y-H, Huybrechts KF, Patorno E, et al. Metformin use in the first trimester of pregnancy and risk for nonlive birth and congenital malformations: emulating a target trial using real-world data. Ann Intern Med 2024;177:862–870
160. Gilbert C, Valois M, Koren G. Pregnancy outcome after first-trimester exposure to metformin: a meta-analysis. Fertil Steril 2006;86: 658–663
161. Barbour LA, Feig DS. Metformin for gestational diabetes mellitus: progeny, perspective, and a personalized approach. Diabetes Care 2019;42: 396–399
162. Barbour LA, Scifres C, Valent AM, et al. A cautionary response to SMFM statement: pharmacological treatment of gestational diabetes. Am J Obstet Gynecol 2018;219:367.e1–367.e7
163. Chew EY, Mills JL, Metzger BE, et al. Metabolic control and progression of retinopathy. The Diabetes in Early Pregnancy Study. National Institute of Child Health and Human Development Diabetes in Early Pregnancy Study. Diabetes Care 1995;18:631–637
164. Sarvepalli SM, Bailey BA, D'Alessio D, et al. Risk factors for the development or progression of diabetic retinopathy in pregnancy: meta-analysis and systematic review. Clin Exp Ophthalmol 2023; 51:195–204
165. Duckitt K, Harrington D. Risk factors for pre-eclampsia at antenatal booking: systematic review of controlled studies. BMJ 2005;330:565
166. Barry MJ, Nicholson WK, Silverstein M, et al.; US Preventive Services Task Force. Screening for hypertensive disorders of pregnancy: U.S. Preventive Services Task Force final recommendation statement. JAMA 2023;330:1074–1082
167. Henderson JT, Whitlock EP, O'Conner E, Senger CA, Thompson JH, Rowland MG. *Low-Dose Aspirin for the Prevention of Morbidity and Mortality From Preeclampsia: A Systematic Evidence Review for the US Preventive Services Task Force*. Rockville, MD, Agency for Healthcare Research and Quality, 2014
168. Roberge S, Bujold E, Nicolaides KH. Aspirin for the prevention of preterm and term preeclampsia: systematic review and metaanalysis. Am J Obstet Gynecol 2018;218:287–293
169. Rolnik DL, Wright D, Poon LC, et al. Aspirin versus placebo in pregnancies at high risk for preterm preeclampsia. N Engl J Med 2017;377: 613–622
170. Hoffman MK, Goudar SS, Kodkany BS, et al.; ASPIRIN Study Group. Low-dose aspirin for the prevention of preterm delivery in nulliparous women with a singleton pregnancy (ASPIRIN): a randomised, double-blind, placebo-controlled trial. Lancet 2020;395:285–293
171. Werner EF, Hauspurg AK, Rouse DJ. A cost-benefit analysis of low-dose aspirin prophylaxis for the prevention of preeclampsia in the United States. Obstet Gynecol 2015;126:1242–1250
172. Zen M, Haider R, Simmons D, et al. Aspirin for the prevention of pre-eclampsia in women with pre-existing diabetes: systematic review. Aust N Z J Obstet Gynaecol 2022;62:12–21
173. Voutetakis A, Pervanidou P, Kanaka-Gantenbein C. Aspirin for the prevention of pre-eclampsia and potential consequences for fetal brain development. JAMA Pediatr 2019;173: 619–620
174. Tita AT, Szychowski JM, Boggess K, et al.; Chronic Hypertension and Pregnancy (CHAP) Trial Consortium. Treatment for mild chronic hypertension during pregnancy. N Engl J Med 2022;386: 1781–1792
175. Harper LM, Kuo HC, Boggess K, et al. Blood pressure control in pregnant patients with chronic hypertension and diabetes: should <130/80 be the target? Am J Obstet Gynecol 2025; 232:482–e481-482 e488
176. American College of Obstetricians and Gynecologists. Clinical guidance for the integration of the findings of the Chronic Hypertension and Pregnancy (CHAP) study, 2022. Accessed 15 Aug 2025. Available from https://www.acog.org/clinical/clinical-guidance/practice-advisory/articles/2022/04/clinical-guidance-for-the-integration-of-the-findings-of-the-chronic-hypertension-and-pregnancy-chap-study
177. ACOG Practice Bulletin No. 203: Chronic Hypertension in Pregnancy. Obstet Gynecol 2019; 133:e26–e50
178. Bateman BT, Patorno E, Desai RJ, et al. Angiotensin-converting enzyme inhibitors and the risk of congenital malformations. Obstet Gynecol 2017;129:174–184
179. Garovic VD, Dechend R, Easterling T, et al.; American Heart Association Council on Hypertension; Council on the Kidney in Cardiovascular Disease, Kidney in Heart Disease Science Committee; Council on Arteriosclerosis, Thrombosis and Vascular Biology; Council on Lifestyle and Cardiometabolic Health; Council on Peripheral Vascular Disease; and Stroke Council. Hypertension in pregnancy: diagnosis, blood pressure goals, and pharmacotherapy: a scientific statement from the American Heart Association. Hypertension 2022;79:e21–e41
180. Chang J-C, Chen Y-J, Chen I-C, Lin W-S, Chen Y-M, Lin C-H. Perinatal outcomes after statin exposure during pregnancy. JAMA Netw Open 2021;4:e2141321
181. Achong N, Duncan EL, McIntyre HD, Callaway L. Peripartum management of glycemia in women with type 1 diabetes. Diabetes Care 2014;37:364–371
182. Riviello C, Mello G, Jovanovic LG. Breastfeeding and the basal insulin requirement in type 1 diabetic women. Endocr Pract 2009;15:187–193
183. Centers for Disease Control and Prevention. Unintended Pregnancy. 2024. Accessed 15 August 2025. Available from https://www.cdc.gov/reproductive-health/hcp/unintended-pregnancy/index.html
184. Victora CG, Bahl R, Barros AJD, et al.; Lancet Breastfeeding Series Group. Breastfeeding in the 21st century: epidemiology, mechanisms, and lifelong effect. Lancet 2016;387:475–490
185. Stuebe AM, Rich-Edwards JW, Willett WC, Manson JE, Michels KB. Duration of lactation and incidence of type 2 diabetes. JAMA 2005;294: 2601–2610
186. Pereira PF, Alfenas RdCG, Araújo RMA. Does breastfeeding influence the risk of developing diabetes mellitus in children? A review of current evidence. J Pediatr (Rio J) 2014;90:7–15
187. Pathirana MM, Ali A, Lassi ZS, Arstall MA, Roberts CT, Andraweera PH. Protective Influence of Breastfeeding on Cardiovascular Risk Factors in Women With Previous Gestational Diabetes Mellitus and Their Children: a Systematic Review and Meta-Analysis. J Hum Lact 2022;38:501–512
188. Noctor E, Crowe C, Carmody LA, et al.; ATLANTIC-DIP investigators. Abnormal glucose tolerance post-gestational diabetes mellitus as defined by the International Association of Diabetes and Pregnancy Study Groups criteria. Eur J Endocrinol 2016;175:287–297
189. Liu Z, Zhang Q, Liu L, Liu W. Risk factors associated with early postpartum glucose intolerance in women with a history of gestational diabetes mellitus: a systematic review and meta-analysis. Endocrine 2023;82:498–512
190. Waters TP, Kim SY, Werner E, et al. Should women with gestational diabetes be screened at delivery hospitalization for type 2 diabetes? Am J Obstet Gynecol 2020;222:73.e71–73.e11
191. Werner EF, Has P, Rouse D, Clark MA. Two-day postpartum compared with 4- to 12-week postpartum glucose tolerance testing for women with gestational diabetes. Am J Obstet Gynecol 2020;223:439.e1–439.e7
192. Cohen N, Toledano Y, Shafir M, Lavie O, Zilberlicht A. Evaluation of postpartum dysglycemia by serum glycosylated hemoglobin in patients with gestational diabetes mellitus: a national data analysis. Arch Gynecol Obstet 2024;310:3029–3035
193. Vounzoulaki E, Khunti K, Abner SC, Tan BK, Davies MJ, Gillies CL. Progression to type 2 diabetes in women with a known history of gestational diabetes: systematic review and meta-analysis. Bmj 2020;369:m1361
194. Li Z, Cheng Y, Wang D, et al. Incidence rate of type 2 diabetes mellitus after gestational diabetes mellitus: a systematic review and meta-analysis of 170,139 women. J Diabetes Res 2020; 2020:3076463
195. Mussa J, Rahme E, Dahhou M, Nakhla M, Dasgupta K. Incident diabetes in women with patterns of gestational diabetes occurrences across 2 pregnancies. JAMA Netw Open 2024;7:e2410279
196. Tobias DK, Hu FB, Chavarro J, Rosner B, Mozaffarian D, Zhang C. Healthful dietary patterns and type 2 diabetes mellitus risk among women with a history of gestational diabetes mellitus. Arch Intern Med 2012;172:1566–1572
197. Villamor E, Cnattingius S. Interpregnancy weight change and risk of adverse pregnancy outcomes: a population-based study. Lancet 2006; 368:1164–1170
198. Martínez-Hortelano JA, Cavero-Redondo I, Álvarez-Bueno C, Díez-Fernández A, Hernández-Luengo M, Martínez-Vizcaíno V. Interpregnancy weight change and gestational diabetes mellitus: a systematic review and meta-analysis. Obesity (Silver Spring) 2021;29:454–464
199. Dennison RA, Chen ES, Green ME, et al. The absolute and relative risk of type 2 diabetes after gestational diabetes: a systematic review and meta-analysis of 129 studies. Diabetes Res Clin Pract 2021;171:108625
200. Li N, Yang Y, Cui D, et al. Effects of lifestyle intervention on long-term risk of diabetes in women with prior gestational diabetes: a systematic review and meta-analysis of randomized controlled trials. Obes Rev 2021;22:e13122

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201. Ratner RE, Christophi CA, Metzger BE, et al.; Diabetes Prevention Program Research Group. Prevention of diabetes in women with a history of gestational diabetes: effects of metformin and lifestyle interventions. J Clin Endocrinol Metab 2008;93:4774–4779
202. Aroda VR, Christophi CA, Edelstein SL, et al.; Diabetes Prevention Program Research Group. The effect of lifestyle intervention and metformin on preventing or delaying diabetes among women with and without gestational diabetes: the Diabetes Prevention Program Outcomes Study 10-year follow-up. J Clin Endocrinol Metab 2015;100:1646–1653
203. ACOG Committee Opinion No. 741: Maternal Immunization. Obstet Gynecol 2018;131:e214–e217
204. Maternal Immunization Practice Advisory October 2022. Accessed 15 August 2025. Available from https://www.acog.org/clinical/clinical-guidance/practice-advisory/articles/2022/10/maternal-immunization
205. ACOG Clinical Practice Guideline No. 6. Viral Hepatitis in Pregnancy. Obstet Gynecol 2023; 142:745–759

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